

Can Equity Option Returns Be Explained by a Factor Model? IPCA Says Yes

Amit Goyal

Swiss Finance Institute at the University of Lausanne, Switzerland

Alessio Saretto

Federal Reserve Bank of Dallas, United States

A number of delta-hedged equity option strategies exhibit very large average returns. We show that much of the profitability of these strategies can be explained by an IPCA factor model. The economic magnitude of the return-adjustment produced by IPCA is impressive: even before transaction costs, the average IPCA alpha of 46 long-short trading strategies constructed on previously discovered signals, is close to zero and contrasts with average realized returns of over 80 basis points per month. Our IPCA model can be used as a benchmark for assessing the performance of other option portfolios. (*JEL* G11, G12, G13)

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There is an extensive literature on factor models for stocks (see, e.g., [Fama and French 2015](#); [Hou et al. 2021](#)) and a growing literature on factor models for corporate bonds (see, e.g., [Chung, Wang, and Wu 2019](#); [Lin, Wang, and Wu 2011](#); [Kelly, Palhares, and Pruitt 2023](#)). However, despite the proliferation of studies that document portfolio trading strategies in options that seem to generate abnormal profits (we will discuss these studies later in the section on related literature), literature on factor models for equity options is scant. In this paper, we propose an option factor model that can explain a substantial portion of realized returns for most option trading strategies. Expected returns from

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our model are very close to their realized counterparts. Accordingly, the most commonly studied option trading strategies display little abnormal profitability.

We study delta-hedged option returns. In this case, most (if not all) option pricing models show that option returns depend on volatility dynamics of the underlying stock returns, interest rate dynamics, and jump risks. A factor structure of stock returns implies a factor structure for stocks' total volatility.¹ Thus, part of the change in a stock's volatility is due to change in aggregate volatility. Accordingly, the literature often uses the aggregate volatility factor of [Coval and Shumway \(2001\)](#) to explain option returns. At the same time, [Herskovic et al. \(2016\)](#) provide evidence of a factor structure in idiosyncratic volatility. Therefore, it is likely that another part of the change in a stock's volatility is due to change in idiosyncratic volatility and, thus, option returns are related to commonality in idiosyncratic volatility. Option returns are also related to changes in interest rates. [Muravyev, Pearson, and Pollet \(2022\)](#) show that the interest rate used to price options may be driven in part by borrow fees, which differ across stocks. These borrow fees may also have systematic and idiosyncratic components.² Finally, to the extent that there is commonality in jump risk (possibly related to default risk as in [Vasquez and Xiao 2024](#)), option returns may share a common component related to jump and/or higher-order risks. It is, thus, possible that characteristics that explain the cross-section of option returns do so because they proxy for betas on latent common factors related to volatility, borrow rates, and/or jump risks.

Instrumented principal component analysis (IPCA) of [Kelly, Pruitt, and Su \(2019\)](#) is, thus, ideally suited to aid our search for an options factor model. IPCA helps in reducing the dimension of the many characteristics that are related to option returns and allows a parsimonious, yet intuitive, way to model time varying betas. These betas are allowed to be functions of stock, option, and contract characteristics. IPCA eschews the need to construct portfolios on an ad-hoc basis and, instead, performs a dimension reduction of many individual assets into a few latent factors. Therefore, IPCA can be used to extract factors from individual option returns potentially including multiple option contracts on the same underlying stock.

In our empirical analysis, we construct a library of predictor variables for option returns over the sample period from 1996 to 2022. Our characteristics come from previous studies, such as [An et al. \(2014\)](#), [Boyer and Vorkink \(2014\)](#), [Bali and Murray \(2013\)](#), [Cao and Han \(2013\)](#), [Christoffersen et al. \(2018\)](#), [Goyal and Saretto \(2009\)](#), [Hu and Jacobs \(2020\)](#), [Vasquez \(2017\)](#), and [Zhan et al. \(2022\)](#), and can be classified into four categories: (1) contract characteristics, such as maturity, open interest, and option Greeks; (2) descriptors of risk-neutral return distribution, such as model-free implied

¹ If the returns are governed by a factor structure, such as $R_{it} = \beta_i' F_t + \epsilon_{it}$, then the variance of the stock return is given by $\sigma_i^2 = \beta_i' \Sigma_F \beta_i + \sigma_{\epsilon_i}^2$. Furthermore, the covariance of returns is given by $\sigma_{ij} = \beta_i' \Sigma_F \beta_j + \sigma_{\epsilon_i \epsilon_j}$ (the last term may be nonzero because the residuals may share commonality, à la [Herskovic et al. 2016](#)).

² While option Greeks capture sensitivity of option prices to some of these risks, these Greeks are not option "betas" to option risk factors because they measure sensitivities to stock-specific risks only.

volatility, skewness, and kurtosis; (3) measures of physical return distribution, such as realized volatility, skewness, kurtosis, and autocorrelation; and (4) firm-level characteristics, such as size, price, and book-to-market. In total, we use 46 characteristics as predictors of option returns and as IPCA conditioning variables.

We study expiration to expiration returns defined as the percentage change between an option position from Monday after expiration till expiration Friday next month. Thus, the option contract at the initiation of the position has residual maturity of around 25 days. The option positions are delta-hedged daily up to the contract expiration. We show that many predictors generate statistically significant average returns to long-short delta-hedged call option portfolios obtained by sorting stocks based on the set of 46 characteristics. A few predictors produce extremely significant average returns with t -statistics larger than 10. For example, the average monthly return from sorts on the Goyal and Saretto (2009) realized minus implied volatility spread (RV–IV) is 2.87% (t -statistic = 24.51).

Because it is well-known that transaction costs in option markets are high (see, e.g., Christoffersen et al. 2018; Muravyev 2016), we also calculate portfolio returns net of transaction costs, calculated following Muravyev and Pearson (2020). Note that our expiration to expiration portfolios incur option transaction costs only at the initiation leg of the portfolio. Even after controlling for option transaction costs, we find significant returns for 16 strategies. For example, the net average return to the RV–IV strategy is 1.73% (t -statistic = 14.94). Our findings, thus, largely confirm results in earlier studies of option strategies. These studies often propose risk-adjustments using factor models from stocks and find large alphas from such factor models. Thus, the profitability of option strategies suggests that option markets exhibit huge inefficiencies.

Our contention is that this conclusion is premature and most likely a function of the factor models. While the joint hypothesis problem afflicts all empirical studies, it is possible that the extant factor models exacerbate this issue because option returns are highly nonlinear and should have dynamic factor loadings. Our IPCA factor model is designed to ameliorate these problems by extracting latent factors with dynamic betas from individual option returns.

In estimating the IPCA model, we vary the number of factors from one to five but find that three factors are sufficient to explain delta-hedged call option returns. For example, our baseline 3-factor IPCA model produces time-series R^2 (cross-sectional average of time-series R^2 for each delta-hedged call position) of 15.68% and cross-section R^2 (time-series average of the cross-sectional R^2 for each month) of 11.38%. These R^2 numbers are nontrivial and compare favorably with similar numbers reported for stocks (Kelly, Pruitt, and Su 2019), corporate bonds (Kelly, Palhares, and Pruitt 2023), and cryptocurrencies (Babiak and Bianchi 2023). When we consider the pricing performance of managed portfolios (i.e., zero-cost portfolios that weight individual positions based on their ranks, transformed into a z -score)

constructed on the same 46 characteristics that we use in our IPCA, these R^2 s are around 75%.

One concern with IPCA models is that the factors and model parameters are estimated using the full-sample. While unconditional betas (or parameters of conditional betas) are estimated using full sample in most empirical studies, the extraction of latent factors could raise doubts about data overfitting. To allay these concerns, we also perform an out-of-sample analysis. In this analysis, we estimate latent factors and model parameters on an expanding window basis and use these estimates to make real-time forecasts for the subsequent month. We find only a modest decrease in out-of-sample R^2 relative to their in-sample counterparts for managed portfolios. For example, for the case of 3-factor model, the cross-section R^2 is 75.70% in-sample and 69.62% out-of-sample. As in [Kelly, Palhares, and Pruitt \(2023\)](#), the reason for this modest deterioration in performance is the relatively few parameters that are estimated in IPCA.

Applying the estimated IPCA model to the list of trading strategies, we find that, accounting for multiple hypothesis testing (based on the 5% FDR of [Benjamini and Hochberg 1995](#)), only two strategies have statistically significant IPCA alphas (five at conventional levels). The magnitude of the alpha, even considering only those that are significant, is much lower compared to the magnitude of the long-short raw returns. For example, the alpha of the RV–IV long-short strategy is only 0.34% (t -statistic = 4.58) compared to the return of 2.87% (t -statistic = 24.51). Notably, no trading strategy has a significant alpha after accounting for transaction costs.

We conclude that the extant evidence on option trading strategies does not present any significant challenge to market efficiency. It is important to bear in mind, however, that IPCA factors are statistically constructed to capture systematic variation in the cross-section of delta-hedged returns. The notion of anomalies and/or market efficiency should, therefore, be viewed merely from the lens of investors who must bear the risk to earn returns to these factors, regardless of their economic origins.

We also analyze how the factors' realized returns behave in periods of particular stress in the market. In particular, we consider (a) negative aggregate shocks to market return and VIX proposed by [Büchner and Kelly \(2022\)](#) and [Karakaya \(2013\)](#), (b) proxies that measure changes in the tails of the aggregate distribution, such as tail risk measure of [Kelly and Jiang \(2014\)](#) and conditional risk premiums of the skewness and kurtosis of S&P 500 return distribution following [Carr and Wu \(2009\)](#) and [Kozhan, Neuberger, and Schneider \(2013\)](#), and (c) states of world related to the role of intermediaries ([Baltussen et al. 2021](#); [He, Kelly, and Manela 2017](#)). We find that the realizations of the first factor, which carries the largest premium at 2.06% per month, are lower when economic conditions are worse. The other two factors, which have average returns of 0.26%, and 0.11% per month, are negatively related to some

economic conditions but appear to hedge others. Overall, three factors span different states of nature characterized by different dimensions of risk.

In analyzing the mapping from characteristics to covariances/dynamic betas, we find an important role of RV–IV. This characteristic is not only important for explaining the returns to the [Goyal and Saretto \(2009\)](#) strategy based on RV–IV but also captures commonality in many other predictors of option returns. When we look at the relation between managed portfolio returns and factor returns, we also find that the main contributor to the IPCA factor returns is the return to the managed portfolio strategy based on RV–IV.

Because we study delta-hedged returns, that essentially represent a volatility position, the large role of RV–IV is not completely unexpected. To the extent that RV–IV proxies for the difference between physical and risk-neutral expectations of volatility, it should be related to the expected return of a delta-hedged position. In turn, in a factor model representation, the expected return is the product of a beta and a risk premium. The fact that we find such a strong relation between RV–IV and IPCA conditional betas is then quite reassuring. An open question remains of why the return to the RV–IV managed portfolio is so strongly related to latent factors that IPCA recovers. We plan to address this issue in future work.

1. Related Literature

One can classify option return predictors as broadly related to signals coming from the underlying stocks (or firms) or those coming from options themselves. Regarding stock predictors of option returns, [Cao and Han \(2013\)](#) show that delta-hedged option returns decrease with the stock idiosyncratic volatility. [Galai and Masulis \(1976\)](#) and [Hu and Jacobs \(2020\)](#) show, both theoretically and empirically, that the expected returns of calls (puts) decrease (increase) with underlying stock volatility. [Aretz, Lin, and Poon \(2023\)](#) show that it is important to distinguish the effect of idiosyncratic and systematic volatility in studying these relations. [An et al. \(2014\)](#) show that implied volatility increases for options on stocks with high past returns. [Ramachandran and Tayal \(2021\)](#) report a monotonic relation between various measures of short-sales constraints and delta-hedged put returns on overpriced stocks. [Zhan et al. \(2022\)](#) show that many stock characteristics (such as profit margin, firm profitability, cash holding, and shares issuance) have a strong predictive power for delta-hedged option returns.

Many studies also use option characteristics to predict option returns. [Boyer and Vorkink \(2014\)](#) find that ex ante option implied skewness predicts option returns negatively. Relatedly, [Bali and Murray \(2013\)](#) find a negative relation between risk-neutral stock skewness and the returns of skewness assets constructed from options. [Byun and Kim \(2016\)](#) report that call options written on the most lottery-like stocks underperform otherwise similar call options written on the least lottery-like stocks. [Christoffersen et al. \(2018\)](#) find a

positive risk-adjusted return spread for illiquid over liquid equity options. [Goyal and Saretto \(2009\)](#) find that the difference between historical volatility and implied volatility predicts delta-hedged and straddle returns. [Heston et al. \(2023\)](#) find that momentum works in option markets too. [Muravyev \(2016\)](#) documents that option market order-flow imbalance significantly predicts daily option returns. [Ruan \(2020\)](#) finds that volatility-of-volatility has predictive power for option returns. [Vasquez \(2017\)](#) shows that difference in long-term and short-term implied volatility has predictive power for delta-hedged option returns. [Bali et al. \(2023\)](#) and [Shafaati, Chance, and Brooks \(2021\)](#) apply machine learning techniques to uncover which characteristics are important for explaining the cross-section of option returns (see also [Goyenko and Zhang 2022](#) for using machine learning to predict the joint cross-section of stock and option returns).

The literature on factor models for option returns is small. [Coval and Shumway \(2001\)](#) propose a factor constructed from zero-beta straddles to explain S&P 500 index option returns. [Büchner and Kelly \(2022\)](#) apply IPCA techniques to construct factors for index options. [Karakaya \(2013\)](#) identifies three factors based on level, slope, and value in individual naked option returns. [Broadie, Chernov, and Johannes \(2009\)](#), however, suggest that straddles or delta-hedged option returns are more immune than naked option returns to sampling problems. More recently, [Horenstein, Vasquez, and Xiao \(2022\)](#) construct option factors by using RP-PCA of [Lettau and Pelger \(2020\)](#). Their factor model presents an improvement over option factors constructed from index options of [Coval and Shumway \(2001\)](#) and [Büchner and Kelly \(2022\)](#). While we view their model as a complementary approach to modeling option returns, IPCA has two advantages over RP-IPCA. First, IPCA can handle information at the firm (and potentially at the option contract) level rather than just at the portfolio level, and second, IPCA allows for dynamic betas.

2. Data

2.1 Sample and variable construction

Options data are from OptionMetrics and cover the period from January 1996 through December 2022. We impose two sets of filters. The first type of filter applies to all observations and is aimed at eliminating contracts that are nonstandard or prices that violate basic no-arbitrage bounds or market minimum specifications. We eliminate contracts (a) for which the underlying is not a share code 10 or 11 stock, (b) that have nonstandard settlements (i.e., not the closing of the expiration day), (c) that have nonstandard expiration (i.e., weeklies), (d) that settle for a number of underlying shares different from 100, or (e) for which one of the bid or ask quotes is missing, (f) for which the ask is lower than the bid, or the ask is \$5 above the bid, (g) the bid-ask spread is lower than the minimum tick size (i.e., 5 cents for option prices below \$3 and 10 cents for option prices above \$3), (h) for which the midpoint price is below

or \$100 above the current exercise payoff, or (i) option delta is less than equal to zero for calls, or greater than equal to zero for puts.

The second set of filters only applies to prices that are used to construct the first leg of the return. We study expiration to expiration returns defined as the percentage change between an option position from Monday after expiration till expiration Friday next month, where the option contract at the initiation of the position has residual maturity of around 25 days.³ For example, for calculation of return for the “month” of February 2022, we initiate the position on Monday January 24, 2022, and terminate the position on Friday February 18, 2022. We apply the following filters only to the prices on January 24.⁴ We consider options (a) that are closest to ATM with moneyness (defined as the ratio of strike to underlying price) between 0.8 and 1.2, (b) that have positive volume or positive open interest, (c) for which the midpoint price is higher than 25 cents, and (d) the percentage bid-ask spread is less than 50%. In addition, to avoid issues related to early exercise of American options, we remove all options on underlying stocks that pay a dividend during the holding period (i.e., between January 24 and February 18 in the example above).

As the literature mostly focuses on daily delta-hedged call returns as the norm for investigating cross-sectional patterns in stock option returns (see, e.g., Cao and Han 2013; Zhan et al. 2022), we also consider daily rebalanced delta-hedged calls.⁵ We define returns to buying a delta-hedged call as:

$$\text{DHC}_{t+1} = \frac{(C_{t+1} - C_t) - \sum_{n=0}^{N-1} \Delta_{d_n} (S_{d_{n+1}} - S_{d_n})}{\Delta_t S_t - C_t} - R_{f,t+1}, \quad (1)$$

where we rebalance the hedge at each of the days $d_n, n=0, \dots, N-1$, $d_0=t$, $d_N=t+1$, $C_{t+1}=\max(S_{t+1}-K, 0)$, and R_f is the riskfree return. Following standard convention, we scale the delta-hedged gains by $(\Delta_t S_t - C_t)$ to guarantee that the denominator is always positive. Therefore, our return is the negative of the return to writing a delta-hedged call.

Stock-level information is obtained from CRSP/Compustat data, which are matched and aligned in time with OptionMetrics information so that no look ahead biased is introduced. All accounting information is assumed to be available 6 months after the fiscal year-end. For each stock-month return position we construct monthly characteristics that can generally be divided into four categories:

1. Contract characteristics, such as moneyness, bid-ask spread, open interest, volume, option price, option delta, option vega, option gamma, and implied volatility.

³ Initiating the position on Monday allows us to consider option chains that are introduced on Monday and have had some volume/open interest to satisfy our sample construction filters.

⁴ Duarte et al. (2023) highlight issues in which researchers inadvertently apply infeasible ex post filters to the data from the perspective of a real time trading strategy.

⁵ We analyze delta-hedged puts and straddles in Internet Appendix Section IA4.

2. Measures of risk-neutral distribution. Model-free implied volatility, model-free implied skewness, and model-free implied kurtosis are constructed using options with the same moneyness and the same maturity. We also construct variables using the implied volatility surface: the ATM level (of implied volatility) for maturity of 30 days; the slope of the 30 days curve, as the difference between implied volatility for moneyness of 0.8 and the corresponding value for moneyness of 1.0; the ATM term, as the difference between the implied volatility value for 30 days and the corresponding value for 360 days maturity; and the volatility of implied volatility.
3. Measures of the physical distribution of returns, such as the stock returns: volatility, autocorrelation, skewness, kurtosis, cumulative return of the previous 11 months, the largest absolute return in the past 10 days (Max10), turnover, and idiosyncratic volatility.
4. Stock returns predictors that have been found to predict also option returns such as the RSI, BM, asset, debt, leverage, profit margin, profitability, equity issuance, external financing cash, cash flow variance, analyst dispersion, lever of institutional ownership, and Z-score.

We refer to Appendix A for detailed descriptions of the variables constructions and their sources.

2.2 Empirical regularities

We construct equal-weighted portfolios of daily rebalanced delta-hedged call positions by sorting stocks into deciles based on the characteristics detailed in the previous section. Following the bulk of the literature (see, e.g., [Aretz, Lin, and Poon 2023](#); [Bali and Murray 2013](#); [Cao and Han 2013](#); [Goyal and Saretto 2009](#); [Hu and Jacobs 2020](#)), we consider expiration to expiration returns and sort stocks on Monday following expiration Friday (i.e., we do not skip a day between portfolio formation and trading initiation).⁶

For each sorting variable, we calculate the return of long-short portfolios that invest in the extreme deciles. These returns are calculated using midpoint prices (average of bid and ask). To ease interpretation, we construct the trading strategy so that the full-sample average return is positive. In other words, for some variables, we present results of a 1–10 portfolio, while for others we present results of a 10–1 portfolio. The second column of Table 1 shows the construction methodology used for each variable. We refer to these returns as those on long-short strategies. The third column reports average return and related *t*-statistics. To caution against the interpretation of single test

⁶ Some authors analyze option returns constructed from month-end to month-end to align with the extensive literature on stock return predictability (see, e.g., [Zhan et al. 2022](#)). We consider this alternative return construction in Internet Appendix Section IA5.

Table 1
Long-short delta-hedged call portfolios

	Construction	Return		Return w/ TC	
Moneyness	1–10	0.38	(3.60)	–0.86	(–8.13)
Bid-ask spread	1–10	0.71	(8.07)	–0.13	(–1.50)
Open interest	1–10	0.04	(0.66)	–0.59	(–9.84)
Delta	10–1	0.22	(2.58)	–0.68	(–7.52)
Vega	10–1	1.44	(14.83)	0.47	(5.02)
Gamma	1–10	0.67	(7.09)	–0.25	(–2.69)
Volume	10–1	0.23	(3.11)	–0.40	(–5.28)
Option price	10–1	0.65	(5.72)	–0.12	(–1.09)
IV ATM	1–10	2.34	(17.94)	1.40	(10.95)
IV slope	10–1	2.15	(25.53)	0.91	(10.46)
IV term	10–1	1.42	(15.94)	0.43	(4.84)
IV vol	1–10	1.45	(15.65)	0.61	(6.81)
MFvol	1–10	1.16	(5.16)	0.69	(3.13)
MFskew	1–10	0.83	(4.98)	0.42	(2.54)
MFkurt	10–1	0.74	(4.41)	0.33	(1.97)
Stock return	10–1	0.04	(0.46)	–0.93	(–9.30)
Stock return I I	10–1	0.29	(2.91)	–0.71	(–7.09)
RV	1–10	1.06	(9.23)	0.22	(1.94)
Rskew	1–10	0.14	(2.23)	–0.74	(–12.62)
Rkurt	1–10	0.91	(14.48)	0.15	(2.45)
Turnover	1–10	0.13	(1.32)	–0.66	(–6.68)
IdiosynVol	1–10	1.15	(11.18)	0.33	(3.26)
Max10	1–10	0.94	(8.35)	0.11	(0.99)
Autocorrelation	10–1	0.07	(1.24)	–0.72	(–12.25)
RV–IV	10–1	2.87	(24.51)	1.73	(14.94)
RV–MFvol	10–1	1.64	(7.49)	1.11	(5.16)
Rskew–MFskew	1–10	0.11	(0.73)	–0.31	(–2.13)
Rkurt–MFkurt	1–10	1.00	(8.03)	0.65	(5.22)
BM	10–1	0.13	(1.39)	–0.75	(–8.33)
Profitability	10–1	0.92	(11.71)	0.03	(0.41)
InstOwn	10–1	0.93	(12.10)	0.03	(0.38)
MarketCap	10–1	1.59	(16.74)	0.67	(7.22)
RSI	10–1	0.10	(1.56)	–0.63	(–9.77)
Assets	10–1	1.46	(14.59)	0.64	(6.71)
Debt	10–1	0.68	(8.82)	0.04	(0.50)
Leverage	10–1	0.00	(0.06)	–0.84	(–15.11)
CashFlowVar	1–10	0.72	(9.11)	–0.01	(–0.16)
Cash to asset	1–10	0.87	(9.42)	–0.04	(–0.43)
AnalystDisp	10–1	0.52	(8.18)	–0.49	(–7.82)
1yr NewIss	1–10	0.43	(5.58)	–0.34	(–4.45)
5yr NewIss	1–10	0.65	(8.33)	–0.09	(–1.25)
Profit margin	10–1	1.15	(13.37)	0.25	(3.05)
Stock price	10–1	1.73	(16.09)	0.74	(7.03)
ROE	10–1	0.82	(11.09)	–0.07	(–0.99)
ExternalFin	1–10	0.52	(6.10)	–0.31	(–3.70)
Z-score	10–1	0.68	(7.99)	–0.29	(–3.62)

The table reports average returns (in percent per month) with and without accounting for transaction costs of 46 long-short portfolios. We construct the trading strategies so that average returns are positive by either buying options positions in decile 10 and writing option positions in decile 1, or by buying options in decile 1 and writing options in decile 10. We tabulate the portfolio construction in the second column. Transaction costs are calculated using a ratio of effective to quoted spread equal to 30%. *t*-statistics larger than the [Benjamini and Hochberg \(1995\)](#) 5% FDR adjusted threshold (i.e., 2.25 for mean return and 2.44 for transaction cost adjusted returns) are reported in bold. The sample period is 1996 to 2022.

statistical significance, we provide multiple hypothesis testing (MHT) adjusted significance accounting for a 5% FDR following [Benjamini and Hochberg \(1995\)](#) (for a review of multiple hypothesis methods, see [Harvey, Liu, and Saretto 2020](#)).

Table 1 shows that many characteristics are strong predictors of delta-hedged call returns. Of the 46 predictors, 39 have statistically significant average returns to long-short portfolios (MHT adjustment versus conventional level does not change this number). A few predictors produce extremely significant average returns with t -statistics larger than 10. For example, the average monthly return resulting from sorts on RV–IV, IV ATM, and IV slope is 2.87%, 2.34%, and 2.15%, respectively, with t -statistics of around 20. Relative to the Black-Scholes idealized framework of expected excess return on delta-hedged call return of zero, the magnitude of many of these returns is large.⁷

It is well-known that transaction costs in option markets are high (see, e.g., Christoffersen et al. 2018; Muravyev 2016; Muravyev and Pearson 2020). If investors cannot profit from option strategies after accounting for transaction costs, then one may not learn much from factor models that price options in the absence of transaction costs. Therefore, we pay special attention to transaction costs in our analysis. Our empirical choice of studying expiration to expiration returns is already a first step in this direction as our portfolios incur option transaction costs only at the initiation leg of the portfolio.

We adjust the option initial prices to account for transaction costs following the evidence presented by Muravyev and Pearson (2020). We consider a ratio of effective to quoted spread, ESPR/QSPR, equal to 30%. We obtain effective bid and ask prices at which an investor could either buy or sell a contract. For example, if the bid price of an option is \$3 and the ask price is \$4, then a ratio of ESPR/QSPR = 30% implies that we buy the option at \$3.65 for a long position or sell the option at \$3.35 for a short position (instead of the midpoint price of \$3.50). Net returns are reported in the fifth column of Table 1. Since, by construction, the gross returns to all strategies are positive, a negative net return imply that the strategy is not profitable after transaction costs. We find that even net of transaction cost 16 trading strategies still have significant returns. For example, the average monthly net return of the RV–IV strategy is still 1.73% (t -statistic = 14.94).

Can these returns be explained by a factor model? We will turn to answering this question next.

3. IPCA

In this section, we present a detailed description of the IPCA method introduced by Kelly, Pruitt, and Su (2019). Returns on N assets over time period $t = 1, \dots, T$ are assumed to be generated by the model:

$$R_{i,t+1} = \alpha_{i,t} + \beta'_{i,t} F_{t+1} + E_{i,t+1} = (Z'_{i,t} \Gamma_{\alpha}) + (Z'_{i,t} \Gamma_{\beta}) F_{t+1} + E_{i,t+1}, \quad (2)$$

⁷ For some characteristics, the magnitude of returns in Table 1 is different from that in the original studies (for instance, Christoffersen et al. 2018; Zhan et al. 2022). Two reasons explain this difference. One, we study expiration to expiration returns while some studies use month-end to month-end returns. Second, Duarte et al. (2023) raise sample selection issues with some studies. As discussed in Section 2.1, our sample does not have any look-ahead bias.

where F 's are K latent factors and E 's are the residuals. $\alpha_{i,t}$ and $\beta_{i,t}$ are asset specific time-varying intercept and factor loadings that are both assumed to be linearly related to L observable characteristics summarized in $Z_{i,t}$ (including a constant). The $L \times 1$ vector Γ_α and the $L \times K$ matrix Γ_β define the mapping from a potentially large number of characteristics to the mispricing and a small number of risk factor exposures. The parameters Γ are global in the sense that are assumed to be constant across all stocks and all time periods.

Kelly, Pruitt, and Su (2019) emphasize that the estimation of Γ_β amounts to finding a few linear combinations of candidate characteristics that best describe the latent factor loading structure. Thus, IPCA eschews the need to construct portfolios on an ad-hoc basis and instead performs a dimension reduction of many individual assets into a few latent factors.

Denote R_{t+1} as $N_{t+1} \times 1$ vector, Z_t as a $N_{t+1} \times L$ matrix, α_t as a $N_{t+1} \times 1$ vector, β_t as a $N_{t+1} \times K$ matrix. Here N_{t+1} is the number of stocks with a valid return at time $t+1$ and all observable characteristics at time t . Then, we can rewrite the model in matrix form as:

$$R_{t+1} = \alpha_t + \beta_t F_{t+1} + E_{t+1} = (Z_t \Gamma_\alpha) + (Z_t \Gamma_\beta) F_{t+1} + E_{t+1}. \quad (3)$$

It will be useful to define a $L \times L$ matrix $W_t = Z_t' Z_t / N_{t+1}$ and a $L \times 1$ vector $X_{t+1} = Z_t' R_{t+1} / N_{t+1}$. Since, as we describe later in Section 4, we convert the characteristics into ranks and then transform it into a z-score, we can interpret X_{t+1} as the $t+1$ returns on a set of L managed portfolios. The return on the l th portfolio is a weighted average of asset returns with weights determined by the value of l th characteristic for each asset at time t (normalized by the number of nonmissing observations each month, N_{t+1}).

The first order conditions for the estimation of the system are given by:

$$\begin{aligned} \hat{F}_{t+1} &= \left(\hat{\Gamma}_\beta' W_t \hat{\Gamma}_\beta \right)^{-1} \hat{\Gamma}_\beta' (X_{t+1} - W_t \hat{\Gamma}_\alpha) \\ \text{vec}(\hat{\Gamma}') &= \left(\sum_{t=1}^{T-1} W_t \otimes \hat{F}_{t+1} \hat{F}_{t+1}' \right)^{-1} \sum_{t=1}^{T-1} X_{t+1} \otimes \hat{F}_{t+1}, \end{aligned} \quad (4)$$

where $\tilde{F}_{t+1} = [F_{t+1} : 1]'$, and $\Gamma = [\Gamma_\beta : \Gamma_\alpha]$. As Kelly, Pruitt, and Su (2019) show, the first order conditions involve OLS regressions only and, therefore, the estimation proceeds very quickly. As is evident from Equation (4), the estimation requires only the L -dimensional objects W_t and X_{t+1} rather than original N -dimensional asset returns. A further advantage of this method, thus, is the ease with which missing data are handled.

Identifying a unique set of parameters is important in latent factor models, as they are identified only up to a rotation. Models $\Gamma_\beta F$ and $\Gamma_\beta P P^{-1} F$ are identical for any rotation matrix P . Following Kelly, Pruitt, and Su (2019), we impose the normalization that $\Gamma_\beta' \Gamma_\beta$ is the identity matrix, that the unconditional second moment matrix of F is diagonal with descending

diagonal entries, and that the time-series average of F is positive. In models with unrestricted α , we impose an additional identification assumption of $\Gamma'_\alpha \Gamma_\beta = 0$. The identification assumptions do not restrict the model's ability to explain returns, but merely serve as a way to pin down unique parameters of the model. We initialize the algorithm by setting Γ_β equal to the first K eigenvectors of the sample second moment of the managed portfolio returns and factors F to be the first K principal components of the managed portfolio panel. We iterate the two first-order conditions in Equation (4) until convergence. Since $Z'_t Z_t$ is not too volatile, the initial conditions are a good starting guess and convergence is achieved rapidly.⁸

We also estimate a restricted IPCA model where $\Gamma_\alpha = 0$ to test whether risk compensation in returns solely arises from exposure to systematic factors, or whether returns partially line up with characteristics directly, hence constituting compensation without risk.

3.1 Statistical inference

3.1.1 Γ_α and number of factors. We first estimate the unrestricted model to obtain parameters $\hat{\Gamma}_\alpha$ and $\hat{\Gamma}_\beta$ and the time series of the latent factors $\hat{F}_t$. Note that Equation (3) and the definitions of W_t and X_{t+1} imply that:

$$X_{t+1} = W_t \Gamma_\alpha + W_t \Gamma_\beta F_{t+1} + Z'_t E_{t+1} / N_{t+1}. \quad (5)$$

Define the residuals of managed portfolio returns as $U_{t+1} \equiv Z'_t E_{t+1} / N_{t+1}$ and their fitted values as $\hat{U}_{t+1} = X_{t+1} - W_t \hat{\Gamma}_\alpha - W_t \hat{\Gamma}_\beta \hat{F}_{t+1}$. The returns on managed portfolio for the b th bootstrap are generated by:

$$X_{t+1}^b = W_t \hat{\Gamma}_\beta \hat{F}_{t+1} + q_{t+1}^b \hat{U}_{t+1}^b, \quad (6)$$

by imposing the null of zero α ($\Gamma_\alpha = 0$). In Equation (6), $\hat{U}_{t+1}^b$ is the estimated residual from a random time period drawn uniformly from the set of all possible dates and q_{t+1}^b is a Student- t random variable with unit variance and five degrees of freedom, as in Kelly, Pruitt, and Su (2019). Using the bootstrapped sample, we reestimate the unrestricted IPCA model and record the estimated parameter $\hat{\Gamma}_\alpha^b$. Recall that since the estimation of IPCA model requires only W and X matrices, we do not need to bootstrap the asset returns themselves. Finally, we calculate Wald statistics $W_\alpha = \hat{\Gamma}'_\alpha \hat{\Gamma}_\alpha$ in the data and $W_\alpha^b = \hat{\Gamma}'_\alpha^b \hat{\Gamma}_\alpha^b$ in the b th bootstrap. Inferences are drawn by calculating a p -value as the fraction of bootstrapped W_α^b statistics that exceed the value of W_α from the actual data. We use $B = 1,000$ repetitions in the bootstrap.

3.1.2 Characteristic importance. We modify the procedure described above to test significance of individual characteristic. Since $\hat{\Gamma}_\beta = (\hat{\gamma}_{\beta,1}, \dots, \hat{\gamma}_{\beta,l-1},$

⁸ We thank Seth Pruitt for providing code for IPCA estimation on his website.

$\hat{\gamma}_{\beta,l}, \hat{\gamma}_{\beta,l+1}, \dots, \hat{\gamma}_{\beta,L})'$, we can test for the significance of the l th characteristic by setting $\hat{\gamma}_{\beta,l} = 0$, defining $\hat{\Gamma}_{\beta}^l = (\hat{\gamma}_{\beta,1}, \dots, \hat{\gamma}_{\beta,l-1}, 0_{K \times 1}, \hat{\gamma}_{\beta,l+1}, \dots, \hat{\gamma}_{\beta,L})'$, and then bootstrapping managed portfolio returns as

$$X_{t+1}^b = W_t \hat{\Gamma}_{\beta}^l \hat{F}_{t+1} + q_{t+1}^b \hat{U}_{t+1}^b. \quad (7)$$

We calculate Wald statistics $W_{\beta,l} = \hat{\gamma}_{\beta,l}' \hat{\gamma}_{\beta,l}$ in the data and $W_{\beta,l}^b = \hat{\gamma}_{\beta,l}^{b'} \hat{\gamma}_{\beta,l}^b$ in the b th bootstrap. Inferences are drawn by calculating a p -value as the fraction of bootstrapped $W_{\beta,l}^b$ statistics that exceed the value of $W_{\beta,l}$ from the actual data.

3.2 Pricing metrics

We calculate the pricing performance of the model by calculating a variety of R^2 's and pricing error, following Kelly, Palhares, and Pruitt (2023) as follows:

$$\begin{aligned} \text{Total } R^2 &= 1 - \frac{\sum_{i,t} (R_{i,t+1} - Z'_{i,t} \hat{\Gamma}_{\beta} \hat{F}_{t+1})^2}{\sum_{i,t} R_{i,t+1}^2} \\ \text{Time-series } R^2 &= \frac{1}{\sum_i T_i} \sum_i T_i R_i^2; \text{ where } R_i^2 = 1 - \frac{\sum_t (R_{i,t+1} - Z'_{i,t} \hat{\Gamma}_{\beta} \hat{F}_{t+1})^2}{\sum_t R_{i,t+1}^2} \\ \text{Cross-section } R^2 &= \frac{1}{T} \sum_t R_t^2; \text{ where } R_t^2 = 1 - \frac{\sum_i (R_{i,t+1} - Z'_{i,t} \hat{\Gamma}_{\beta} \hat{F}_{t+1})^2}{\sum_i R_{i,t+1}^2} \\ \text{Relative pricing error} &= \frac{\sum_i \alpha_i^2}{\sum_i \hat{R}_i^2}; \text{ where } \alpha_i = \frac{1}{T_i} \sum_t (R_{i,t+1} - Z'_{i,t} \hat{\Gamma}_{\beta} \hat{F}_{t+1}). \end{aligned} \quad (8)$$

Note that the denominator in the total R^2 is the sum of squared returns without demeaning following Gu, Kelly, and Xiu (2020). Total R^2 measures how well the model describes realized returns in a panel context. The time-series R^2 summarizes pricing performance as the cross-sectional average of the performance of each test asset i (we weight each asset's performance by the number of time-series observations for that asset, T_i). The cross-section R^2 quantifies the cross-sectional predictive performance as in a Fama and MacBeth (1973) context. Relative pricing error analyzes IPCA's ability to price average return. In contrast to R^2 measures, lower value of relative pricing error indicates better performance.

Following Kelly, Pruitt, and Su (2019), we also calculate R^2 and relative pricing error in an out-of-sample context. The exact procedure is as follows. For each month t , we first estimate IPCA on expanding window that includes data up to month t . We can write the resultant parameter estimates as $\hat{\Gamma}_{\beta,t}$. Then, we calculate the out-of-sample factor return at time $t+1$ as $\hat{F}_{t+1,t} = (\hat{\Gamma}_{\beta,t}' Z_t' Z_t \hat{\Gamma}_{\beta,t})^{-1} \hat{\Gamma}_{\beta,t}' X_{t+1} = \hat{\theta}_t' X_{t+1}$. In this way, IPCA factor return at time $t+1$

uses portfolio weights, $\hat{\theta}_t$ ($L \times K$ matrix), observable at time t only. The out-of-sample measures in Equation (8) are then calculated by replacing $\hat{\Gamma}_\beta$ with $\hat{\Gamma}_{\beta,t}$ and $\hat{F}_{t+1}$ with $\hat{F}_{t+1,t}$.

4. IPCA Performance

We include all the stock- and option-level characteristics described in Section 2.1 in our baseline results. As is standard in IPCA analysis, we convert all characteristics to cross-sectional ranks from 0 to 1, and then take a normal inverse cumulative distribution function to convert these to z -scores (roughly between -3 and 3).⁹ The advantage of normal inverse transformation is that, relative to ranks, it increases the weights of observations in the tails. This is important in our setting because decile returns to the strategies studied in Section 2.2 are monotonic for the most part but deliver extreme returns in the tails (see Internet Appendix Table IA1 for details). Using the z -score weights makes the resultant managed portfolio return (X using the notation of Section 3) closer to that of long-short portfolio.

It is unclear how to estimate an IPCA model on returns that are net of transaction cost. IPCA factors are portfolio combinations of managed portfolios. Ex ante, it is not obvious whether a particular managed portfolio will have a positive or a negative position in a particular IPCA factor in a particular month; the sign of the position could be different for different IPCA factors and/or for different months. As such, it is not possible to use net option returns as inputs to the IPCA model. In line with all the studies using the IPCA method (those of Babiak and Bianchi 2023; Büchner and Kelly 2022; Kelly, Palhares, and Pruitt 2023; Kelly, Pruitt, and Su 2019), we sidestep the issue of incorporating transaction costs in the *estimation* of IPCA. We do, however, analyze factor returns and alphas from net of transaction cost returns.

4.1 In-sample

We start our analysis with a constrained IPCA where we force the alpha to be zero ($\Gamma_\alpha=0$). Since theory provides little guidance on the number of factors, we perform the IPCA analysis for up to five latent factors. Table 2 tabulates total, time-series, and cross-section R^2 as well as the relative pricing error from Equation (8) for this specification. We report the statistics for individual equity option position as well as for the managed portfolios.

As is typical of other IPCA applications (see, e.g., Kelly, Palhares, and Pruitt 2023), we find that increasing the number of factors monotonically improves performance measures in-sample. The total R^2 for individual option positions

⁹ There are some variables (e.g., volume) for which we observe the same values of Z across stocks. These observations are assigned the same rank and the transformed weight. For purely programming convenience, we assign a value of zero to missing Z . This has, obviously, no impact on any of the results as the option return, even if available, for a firm with missing Z is not incorporated in the managed portfolio return.

Table 2
IPCA performance

	Number of factors				
	1	2	3	4	5
A. In-sample					
Stock-level option positions:					
Total R^2 (%)	8.77	11.54	12.22	12.75	13.22
Time-series R^2 (%)	12.11	15.04	15.68	15.87	16.16
Cross-section R^2 (%)	8.69	10.84	11.38	11.86	12.27
Relative pricing error	78.46	73.68	72.74	71.80	71.20
Managed portfolios:					
Total R^2 (%)	63.93	82.45	87.60	89.41	91.23
Time-series R^2 (%)	40.65	65.15	73.43	76.48	80.33
Cross-section R^2 (%)	53.45	70.50	75.70	78.38	80.58
Relative pricing error	24.46	0.38	0.23	0.18	0.15
Bootstrap test:					
W_α p -value	.00	.01	.00	.00	.03
Economic test:					
# Unpriced factors	0	0	0	1	3
B. Out-of-sample					
Stock-level option positions:					
Total R^2 (%)	6.70	8.74	9.32	9.77	10.26
Time-series R^2 (%)	0.08	-6.76	-6.15	-9.12	-19.25
Cross-section R^2 (%)	6.65	8.32	8.87	9.28	9.77
Relative pricing error	81.75	77.73	76.99	76.39	75.32
Managed portfolios:					
Total R^2 (%)	54.39	78.23	83.27	85.01	87.65
Time-series R^2 (%)	28.22	57.52	66.42	68.95	73.94
Cross-section R^2 (%)	47.19	63.92	69.62	72.14	75.00
Relative pricing error	48.16	9.47	5.26	6.06	5.16

The table presents performance measures of IPCA models for delta-hedged call returns. We report total, time-series, and cross-section R^2 , as well as the relative pricing error (all in percent) from Equation (8) for individual stock option positions and managed portfolios for the constrained ($\Gamma_\alpha=0$) model. We also report p -values, in decimal units, on the bootstrap Wald test of $\Gamma_\alpha=0$. The last row of the top panel reports number of unpriced factors calculated as the number of estimated IPCA factors whose returns are statistically indistinguishable from zero. In panel B, we report out-of-sample performance measures, obtained by splitting the sample in half and using the first half to estimate the model. We then use estimated factors and conditional betas to produce a 1-month out-of-sample forecast. We then roll the estimation procedure forward 1 month at a time, until the end of the sample. Results are reported for IPCA models with $K=1$ to 5. Conditional betas are estimated for each of the characteristics used in Table 1. The sample period is 1996 to 2022.

increases from 8.77% to 13.22% when going from 1 to 5 latent factors. The time-series R^2 and cross-section R^2 are of roughly the same magnitude as the total R^2 . As is usually the case, we also find that the performance measures are higher for managed portfolios than those for individual options. This result obtains partly because the managed portfolios have much lower cross-sectional and time-series dispersion in returns than individual delta-hedged option returns. For example, the total R^2 for managed portfolios is 63.93% for 1-factor model and increases to 91.23% for 5-factor model.

The relative pricing error is around 70% for individual option positions and remains relatively high even when we increase the number of factors. The relative pricing error is high at 24.46% for managed portfolios for the case of 1 latent factor but declines to around 0.20% when the number of latent

factors is bigger than 2. Overall, we conclude that, in-sample, IPCA is able to explain about 8%–10% of the variability in stock-level option returns and about 80%–90% of the variation in managed portfolios returns.

The IPCA option return performance measures compare well with similar estimates for stocks (Kelly, Pruitt, and Su 2019), corporate bonds (Kelly, Palhares, and Pruitt 2023), and index equity options (Büchner and Kelly 2022). Our measures are a bit lower (lower R^2 s and higher relative pricing error) than those in these studies but are very similar to the respective measures extracted by applying IPCA to cryptocurrencies (Babiak and Bianchi 2023).

Although we tabulate results only for the constrained version, we also estimate an unconstrained IPCA model where we place no restriction on alpha. Following the procedure described in Section 3.1.1, we calculate a Wald test and its bootstrapped p -value for the null hypothesis of zero alpha, and report this in the penultimate row of panel A of Table 2. We find that the p -value is generally low. Thus, we fail to reject the hypothesis of zero alpha for up to 5 factors and conclude, from this statistical test, that at least 5 factors are needed to describe the variation in returns with only the time-varying loadings on factors without an intercept.

At the same time, we also notice that IPCA models with higher K do not always result in factors that carry a significant average return. In other words, some of the estimated factors are not priced (have an average return that is statistically insignificantly different from zero). For each IPCA specification, we report the numbers of unpriced factors in the last row of panel A of Table 2. We refer to this as the economic test. For specifications with $K \geq 4$, we always find at least one factor with an average return that is not statistically different from zero. Combining the bootstrap test with the economic test, we conclude that a specification with 3 factors is preferable.

4.1.1 IPCA versus PCA. Given the fact that IPCA extracts latent factors which are ultimately a combination of managed portfolios, one might wonder how the performance of IPCA factors compares to that of factors derived from a traditional PCA. We extract principal components from the set of 47 managed portfolios (i.e., one for each characteristics plus a constant). Conveniently, a visual inspection of the eigenvalues supports the idea that the first three principal components absorbs a large portion of the variability in the data (i.e., the first three eigenvalues capture approximately 85% of the total variance).

We compare the first 3 principal components (henceforth PCA factors), with the 3 IPCA factors of the benchmark model. To get a sense of how the two sets of factors are related to each other, we compute the canonical correlation coefficient (see Boivin and Ng 2006) and find it to be 0.74. In other words, while the two sets of factors are positively related to each other, they seem to span a sufficiently different space.

Second, we run a conditional beta model similar to IPCA where the factors are prespecified as the first 3 PCA factors.¹⁰ The conditional beta PCA model performs relatively well, thus highlighting the importance of conditional betas. However, the performance of the PCA factors is worse than that of the latent IPCA factors, thus highlighting the importance of joint estimation of Γ_β and factors in the IPCA estimation. For example, the Total R^2 for managed portfolios is 85.70% for the PCA factors, versus 87.60% for IPCA. The difference in relative pricing errors is more dramatic: it is 4.53% versus 0.23%, for PCA and IPCA factors, respectively. Increasing the number of factors from 3 to 5 does not improve PCA's performance. For example, even with 5 PCA factors, the relative pricing error is still one order of magnitude larger (at 2.30%) than that of the 3-factor IPCA model.

These results are somewhat in line with those of Kelly, Pruitt, and Su (2019) who find that IPCA does better than PCA with static loadings for stock returns. Our analysis shows that the outperformance of IPCA for delta-hedged option returns carries over even when the PCA factors are allowed to have dynamic loadings.

4.2 Out-of-sample

The results in the previous sections are based on in-sample estimation where we estimate both the factors and the model parameters using the entire sample. This is the usual practice in asset pricing studies where unconditional betas (or parameters of conditional betas in conditional models) are estimated using the full sample. Nevertheless, given that our latent factors are also extracted using full-sample estimation, this could raise doubts about data overfitting. Accordingly, we also consider how the IPCA model performs out-of-sample.

We calculate the statistics from Equation (8) in an out-of-sample fashion as described in Section 3.2. The out-of-sample period is the second half of our sample, from 2009 to 2022 (we use expanding windows in the out-of-sample analysis). The results are tabulated in panel B of Table 2. We find a modest increase in out-of-sample R^2 s to their in-sample counterparts. For example, for $K=3$, the cross-section R^2 for managed portfolios is 75.70% in-sample and 69.62% out-of-sample. Time-Series R^2 have large negative values.¹¹ Pricing errors for individual option positions also change only by modest amount in the out-of-sample test. Although the relative increase is more substantial for

¹⁰ It is easy to adapt the estimation machinery of Section 3 for prespecified factors. See Kelly, Pruitt, and Su (2019) for details. Recall that the IPCA procedure initializes the factors using PCA factors and then computes Γ_β using these factors in the first iteration. Therefore, one can view the conditional beta PCA model as an IPCA model where only Γ_β is updated from one step of the iteration to the other.

¹¹ Computing out-of-sample time-series R^2 for an unbalance panel is challenging. A number of stocks have very few observations. Some of those stocks have realized returns very close to zero, leading to negative R^2_t for those stocks (see Equation (8)). For example, for $K=3$, we report time-series R^2 of -6.15% in Table 2. However, when we remove stocks for which we only have one observation (stocks most likely to have negative out-of-sample R^2_t), the time-series R^2 becomes 12.53% .

managed portfolios, out-of-sample relative pricing errors for portfolios remain low. For example, with 3 latent factors, relative pricing errors are 0.23% and 5.26%, in-sample and out-of-sample, respectively.

As noted before, some of the factors for $K \geq 4$ are statistically weak. It is interesting to see that, out-of-sample, relative pricing error for managed portfolios increases from 3 to 4 factors (but then declines again for 5 factors). This suggests that introducing weak factors has some consequences out-of-sample and provides some more comfort in our choice of $K = 3$ for the number of factors.

It may seem surprising that we do not see much deterioration in out-of-sample performance relative to in-sample performance. As Kelly, Palhares, and Pruitt (2023) argue, this is explained by the relatively few parameters that are estimated in IPCA. With 47 characteristics (including a constant), our three factor model requires 141 conditional beta parameters while a standard static beta model would require more than 100,000 parameter estimates for thousands of stock options.

4.3 Portfolio alphas

We now return to the long-short strategies presented in Table 1. The variables underlying these strategies are the conditioning variables in IPCA. In this section, we discuss how expected option returns extracted from the three factor IPCA model are useful at adjusting the performance of these strategies.

We calculate the average returns and the difference between the realized return and the IPCA expected return (i.e., the IPCA alpha). The expected return of the portfolio is constructed by aggregating the expected return of the individual option positions, at the stock level. For example, if the long leg is composed of 30 stocks in a particular month, then the expected return for that month is the average of the IPCA expected returns of those 30 stocks for the same month. Define the portfolio alpha as

$$\hat{\alpha}_p = \frac{1}{T} \sum_{t=1}^T \left(R_{p,t+1} - Z'_{p,t} \hat{\Gamma}_\beta \hat{F}_{t+1} \right), \quad (9)$$

where $Z_{p,t}$ is the $L \times 1$ vector of the portfolio p 's characteristics at time t (calculated as weighted average of individual stock characteristics). Equation (9) shows that the alpha depends on estimated parameters $\hat{\Gamma}_\beta$ and estimated factors $\hat{F}$. Therefore, one needs to account for the estimation error in these model parameters to properly calculate the standard error of alpha estimates. We provide details on the procedure in Appendix B.

Table 3 reports alphas and t -statistics. The average alpha across the 46 trading strategies is about 6 basis points, or equivalently about 7% of the average raw return. After controlling for MHT, 2 (5 at conventional levels) long-short portfolios still have a statistically significant IPCA alpha: IV slope and RV–IV. These strategies have a raw return of 2.15% and 2.87%,

Table 3
IPCA alphas

Moneyness	−0.05	(−0.52)	Autocorrelation	0.08	(1.34)
Bid-ask spread	−0.01	(−0.08)	RV−IV	0.34	(4.68)
Open interest	0.06	(0.86)	RV−MFvol	−0.27	(−1.24)
Delta	0.09	(0.90)	Rskew−MFskew	−0.04	(−0.24)
Vega	0.08	(0.58)	Rkurt−MFkurt	0.21	(1.51)
Gamma	0.04	(0.28)	BM	0.15	(2.10)
Volume	0.05	(0.65)	Profitability	0.02	(0.25)
Option price	−0.19	(−1.14)	InstOwn	0.03	(0.46)
IV ATM	0.24	(2.11)	MarketCap	0.11	(1.00)
IV slope	0.34	(4.58)	RSI	0.12	(1.71)
IV term	0.16	(2.11)	Assets	0.14	(1.41)
IV vol	0.11	(1.21)	Debt	−0.03	(−0.36)
MFvol	−0.26	(−1.14)	Leverage	−0.01	(−0.15)
MFskew	0.11	(0.63)	CashFlowVar	0.02	(0.18)
MFkurt	−0.15	(−0.82)	Cash to asset	0.08	(1.14)
Stock return	0.13	(1.43)	AnalystDisp	−0.02	(−0.34)
Stock return11	0.03	(0.31)	1yr NewIss	0.09	(1.37)
RV	0.16	(1.41)	5yr NewIss	0.09	(1.34)
Rskew	−0.11	(−1.88)	Profit margin	0.04	(0.53)
Rkurt	0.14	(1.95)	Stock price	0.09	(0.65)
Turnover	0.06	(0.70)	ROE	0.06	(0.79)
IdiosynVol	0.16	(1.65)	ExternalFin	0.04	(0.51)
Max10	0.13	(1.24)	Z-score	0.05	(0.57)

The table presents IPCA alphas (in percent per month) and *t*-statistics in parentheses for long-short portfolios of delta-hedged call returns as described in Table 1. The IPCA model has 3 factors. Portfolios with *t*-statistics larger than the [Benjamini and Hochberg \(1995\)](#) 5% FDR adjusted threshold (i.e., 2.60 and 3.18 for mean return and alphas of long-short portfolios, respectively) are reported in boldface. The sample period is 1996 to 2022.

respectively. Their IPCA expected returns are 1.81% and 2.53%, thus resulting in alphas of 34 basis points. In other words, the model explains more than 84% of the average realized return. From both a statistical and economic point of view, the performance of IPCA is quite remarkable.

We provide a visual exposition by plotting the portfolio alphas against their average realized returns in Figure 1. Alphas that are statistically significant at MHT levels are plotted as red stars, while alphas that are statistically indistinguishable from zero are plotted as blue stars. The figure shows that the IPCA model manages to eliminate most of the alphas, with most of the portfolios located on the horizontal line corresponding to zero alphas (as is already evident from the results in Table 3).

We also examine the alphas of strategies after accounting for transaction costs. Recall that Table 1 shows that 16 strategies still have statistically significant net of transaction cost returns. [Internet Appendix Figure IA1](#) shows that net of transaction cost alphas from the IPCA model are negative for all strategies.

We conduct several robustness checks. These include selecting particular cross-section of options based on liquidity and moneyness, pricing delta-hedged puts and straddles, and pricing month-end to month-end returns. We provide details of all these tests in [Internet Appendix Sections IA2 to IA5](#).

There is an important caveat to our analysis, indeed to the IPCA model. As emphasized by [Kelly, Pruitt, and Su \(2019\)](#), the notion of risk in the statistical

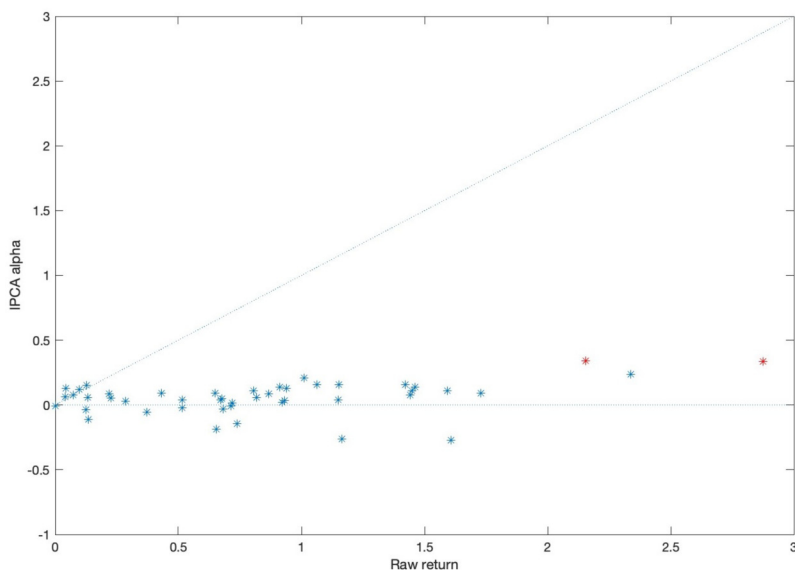


Figure 1
IPCA alphas

The figure plots portfolio alphas (in percent per month) from Table 3 against portfolio returns from Table 1 for long-short portfolios of delta-hedged call returns. The alpha is computed from the IPCA model with 3 factors. The sample period is 1996 to 2022.

ALT TEXT: Relationship between IPCA alphas and raw returns illustrates that IPCA alphas are close to zero for almost all strategies.

IPCA model is that of correlated fluctuations in the cross-section of asset returns arising from common IPCA factors. These factors may not necessarily correspond to those studied in macro-finance or q type models, and may arise from investor sentiment as in, for example, [Kozak, Nagel, and Santosh \(2018\)](#) and [Stambaugh and Yuan \(2016\)](#). Thus, IPCA factors capture nondiversifiable variation in returns regardless of their economic origin. As such, following standard terminology, while we use α as synonymous with anomalous returns, “there can be phenomena considered anomalous from the perspective of a canonical asset pricing theory that would appear as a risk factor in our analysis, so long as those phenomena produce covariation among large swathes of asset returns” [Kelly, Pruitt, and Su \(2019, p. 506\)](#).

5. Interpreting IPCA

While our analysis so far shows that IPCA factors with dynamic conditional betas do well in explaining option returns, one limitation is that these factors are latent. In this section, we seek to understand what IPCA betas and factors might represent.

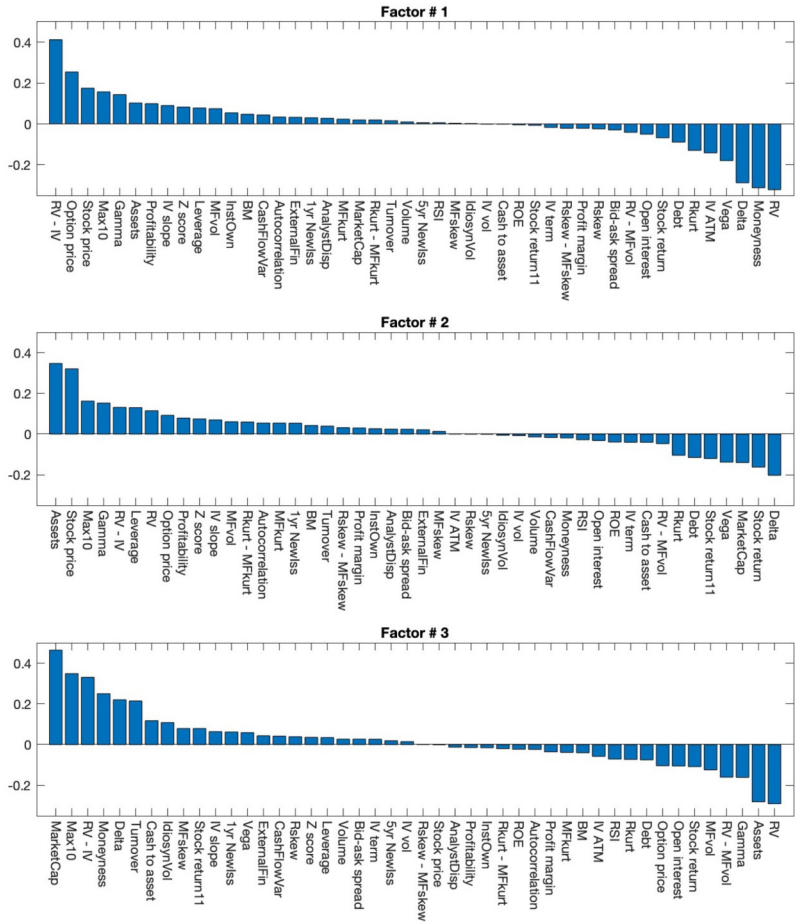


Table 4
RV–IV expected IPCA returns decomposition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ret	−2.89	−1.14	−0.81	−0.59	−0.52	−0.37	−0.25	−0.20	−0.06	−0.02
E(Ret)	−2.41	−1.46	−1.04	−0.75	−0.59	−0.36	−0.22	−0.11	0.02	0.13
α_{IPCA}	−0.48	0.32	0.24	0.16	0.07	−0.01	−0.03	−0.09	−0.07	−0.14
β_{F1} E(F1)	−2.63	−1.70	−1.27	−0.99	−0.84	−0.61	−0.49	−0.40	−0.31	−0.27
β_{F2} E(F2)	0.20	0.20	0.19	0.19	0.19	0.19	0.20	0.20	0.22	0.25
β_{F3} E(F3)	0.02	0.04	0.04	0.04	0.06	0.06	0.07	0.09	0.11	0.15
RV–IV	−1.75	−1.04	−0.68	−0.39	−0.11	0.13	0.39	0.68	1.04	1.75

The table presents a decomposition of the returns (in percent per month) of decile portfolios constructed by sorting stocks based on the difference between realized and implied volatility (RV–IV). Expected return and IPCA alpha are derived from the IPCA model estimated with 3 factors from Table 3. The sample period is 1996 to 2022.

A quick look by singling out the absolute largest coefficient in each panel suggests a large role for RV–IV, Assets, and MarketCap, respectively, for factors 1, 2, and 3. These characteristics are related to expected returns because they proxy for covariances/betas with respect to IPCA factors.

For example, in the context of the IPCA model, options with high RV–IV demand a higher expected return than those with low RV–IV because the former have high loadings on the first factor, everything else equal (the return to the long-short strategy on RV–IV in Table 1 is 2.87%). Stated differently, options with low RV–IV have low expected returns and high prices. Investors are willing to pay higher prices for options with low RV–IV because these securities are hedges (relative to options with high RV–IV) against bad realizations of the first factor.

While Γ_β shows the multivariate dependence of betas on characteristics, it is of interest to know that univariate relation of betas to characteristics. Toward, this end, we calculate beta as $\beta_{i,t} = Z'_{i,t} \Gamma_\beta$ and calculate its cross-sectional correlation with each $Z_{i,t}$ each month. We report the time-series average of the 46 characteristics $\times$ 3 factor betas' correlations in Internet Appendix Figure IA3. We find broad correspondence between Figure 2 and Figure IA3.

To shed some light on how IPCA functions, Table 4 shows several quantities of interest for all the RV–IV decile portfolios. Both realized and IPCA expected returns monotonically increase across the 10 decile portfolios. The IPCA model prices very closely most of the 10 portfolios; only decile 1 has an alpha that fails to line up with the average return. Most of the variation in realized return across the 10 portfolios is addressed by the first factor, where the spread in the betas is strongly related to the spread in RV–IV characteristic itself.

As another example, options on stocks with high Assets have high betas on the second IPCA risk factor, thereby, (at least partly) explaining the returns to the long-short strategy on Assets in Table 1 of 1.46%. However, the mapping from characteristics to betas is not always so straight-forward. For example, IV term does not have a high $\gamma_{\beta,l,k}$ on any of the three factors, yet has a return

on 1.42% in Table 1. In unreported analysis, we find that deciles 1 and 10 of the IV term strategy have low and high betas on the first factor, respectively, because the extreme deciles have large (in absolute terms) values of $RV-IV$. A somewhat similar example is the strategy based on Stock price that earns 1.73% in Table 1. Although Stock price is a relatively important characteristic for the second factor in Figure 2, the variation in expected returns produced by the betas on the second factor is not enough to explain the realized returns. Rather, once again, we find that deciles 1 and 10 of the strategy on Stock price have a spread in betas on the first factor because the extreme deciles have a spread in their $RV-IV$.

Although $RV-IV$ is important for explaining many strategy returns, it is not the only story. An example is provided by the sorts on Profit margin that earn 1.15% in Table 1. Here it is the spread in RV leading to spread in betas on the first factor and to a slightly lesser extent, spread in Assets leading to spread in betas on the second factor that explains the variation in realized returns.

Since the $\gamma_{\beta,l,k}$ coefficient associated with each characteristic changes for each factor (and could even be of different signs across two factors), it is not possible to gauge the overall importance of each characteristic from the previous analysis. Nevertheless, following Kelly, Pruitt, and Su (2019), we compute a statistic that measures the impact of each characteristic by summing up the squares of its gammas (one for each factor) and then taking the square root, $\sqrt{\sum_k \gamma_{\beta,l,k}^2}$. We evaluate the statistical significance of each statistic (there is one for each characteristic) using the bootstrap procedure described in Section 3.1.2. We report the impact statistics and their p -value in Table 5, and present results in descending order of the impact metric (i.e., larger statistic at the top). In general, characteristics with the largest impact metric are also more statistically significant. The characteristics with the larger impact metric that are also statistically significant are $RV-IV$, Assets, and MarketCap. These are the same characteristics that are also visible in Figure 2. Therefore, options with high $RV-IV$, options on stocks with large assets, and options on stocks with high market capitalization, in general, command higher expected returns because these options have higher betas (everything else equal) on IPCA risk factors.

5.2 Which characteristics matter?

So far we have included the same characteristics in the IPCA estimation process as those that are used to construct the long-short strategies that we use to evaluate the model. This raises the question of whether a characteristic needs to be included in the IPCA model for the model to be able to price the associated long-short strategy. We will investigate this next.

5.2.1 Removing insignificant characteristics. While motivated by existing literature, our choice of conditioning characteristic is still arbitrary. On the

Table 5
Characteristics importance

RV–IV	0.54 (.00)	Open interest	0.12 (.02)
MarketCap	0.49 (.00)	Z-score	0.12 (.04)
Assets	0.46 (.00)	IdiosynVol	0.11 (.07)
RV	0.45 (.05)	1yr NewIss	0.09 (.02)
Max 10	0.42 (.00)	MFskew	0.08 (.55)
Delta	0.42 (.03)	RSI	0.08 (.15)
Moneyness	0.40 (.02)	BM	0.08 (.17)
Stock price	0.37 (.08)	MFkurt	0.07 (.60)
Option price	0.29 (.69)	Autocorrelation	0.07 (.02)
Gamma	0.26 (.25)	Rkurt–MFkurt	0.07 (.45)
Vega	0.23 (.39)	CashFlowVar	0.06 (.08)
Turnover	0.22 (.01)	InstOwn	0.06 (.14)
Stock return	0.21 (.04)	ExternalFin	0.06 (.11)
Rkurt	0.18 (.00)	Profit margin	0.05 (.20)
RV–MFvol	0.17 (.36)	IV term	0.05 (.22)
Debt	0.16 (.06)	Bid-ask spread	0.05 (.52)
MFvol	0.16 (.48)	Rskew	0.05 (.16)
Leverage	0.16 (.07)	ROE	0.05 (.20)
IV ATM	0.15 (.45)	AnalystDisp	0.04 (.17)
Stock return11	0.14 (.15)	Rskew–MFskew	0.04 (.69)
IV slope	0.13 (.03)	Volume	0.03 (.06)
Profitability	0.13 (.04)	5yr NewIss	0.02 (.62)
Cash to asset	0.12 (.01)	IV vol	0.02 (.79)

The table presents measures of importance for each characteristic used in the IPCA estimation of Table 2 with 3 factors. Variables are sorted by the square root of the sum of squared Γ . We also report the p -value of the bootstrap test for each $\Gamma_\beta=0$ in parentheses. The sample period is 1996 to 2022.

one hand, more information could be used, but, on the other, some of the information that we use appears to be redundant according to Table 5. Accordingly, we remove all characteristics that are statistically insignificant at the 5% level based on the metric in Table 5 and rerun the IPCA procedure using only the 17 significant characteristics as conditioning variables (plus a constant). We refer to the resultant model as the *restricted* IPCA model.

We evaluate the restricted IPCA model on two grounds: how the model prices single option-level positions, and how the model prices long-short portfolios constructed using the characteristics that are left out of the estimation exercise. On the first count, the Total R^2 goes from 12.22% (third column of Table 2) in the original IPCA model to 11.90% in the restricted IPCA model with 17 characteristics. Similarly, the Relative Pricing Error increases from 72.74% to 74.60%. On the second count, we visualize the results in panel A of Figure 3 by organizing the plot in a similar fashion to Figure 1. Of the 29 (46–17) long-short strategies constructed on characteristics that are left out from the estimation, only 5 remain significant after controlling for MHT. The economic magnitude of the IPCA alpha is small compared to the raw returns: the average IPCA alpha is 4.5 basis points, while the average raw return is 80 basis. Thus, the deterioration (from full IPCA to restricted IPCA) in the ability to price nonincluded strategies is small. This exercise, thus, shows that the characteristics that are significant in Table 5 contain the bulk of the information that matters to explain the cross-section of delta-hedged returns.

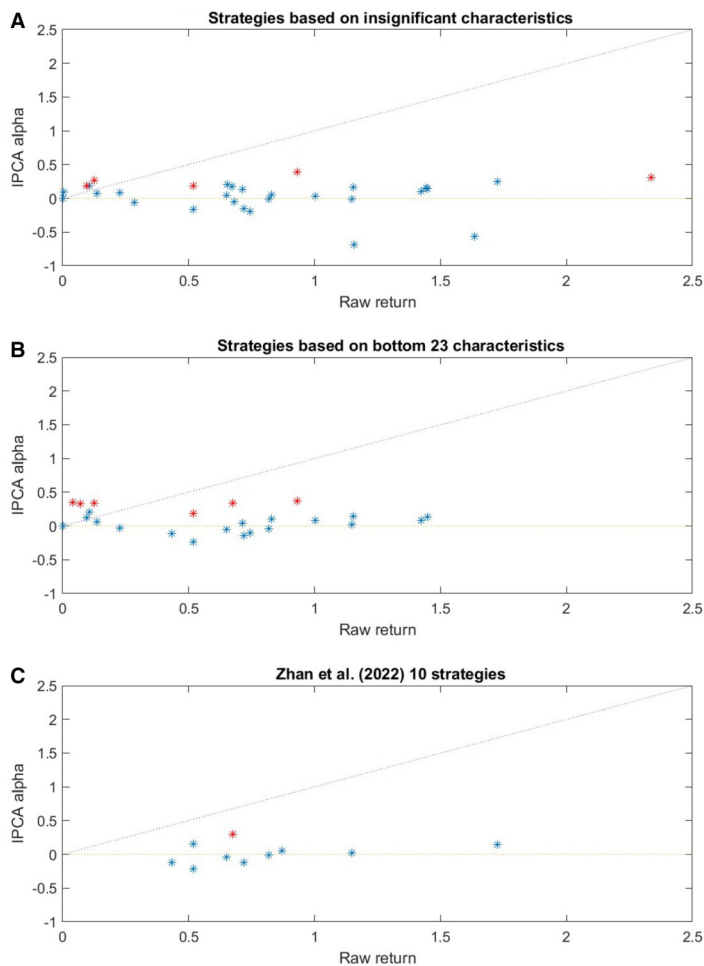


Figure 3
Characteristics subsamples

The figure shows IPCA alphas versus raw returns (both in percent per month) for three different set of characteristics. In each case, we retain some characteristics and estimate a restricted IPCA model which we use to price the long-short portfolios constructed with the characteristics that are not included in the estimation (i.e., target long-short portfolios). In panel A we use the 17 statistically significant characteristics from Table 5 to estimate the model and the remaining 29 statistically insignificant characteristics to construct the target long-short portfolios. In panel B we use the 23 characteristics with the highest impact score from Table 5 to estimate the model and the remaining 23 characteristics to construct the target long-short portfolios. In panel C, we use the 10 characteristics from Zhan et al. (2022) to construct the target portfolios and the remaining 36 to estimate the IPCA model. The sample period is 1996 to 2022.

ALT TEXT: IPCA alphas are close to zero for almost all strategies for different ways of estimating IPCA models (by retaining a subset of characteristics).

5.2.2 Removing unimportant characteristics. Next, we evaluate whether the magnitude of the impact estimates reported in Table 5 matters. This contrasts with the previous exercise where we only considered statistical significance.

We include the 23 most important characteristics (the ones on the left-hand-side of Table 5) in the estimation model and evaluate the strategies constructed based on the remaining 23 characteristics (the ones on the right-hand-side of Table 5). We find that the overall performance of the resultant restricted IPCA model remains strong: the relative pricing error of the restricted IPCA model is 73.90%. We again visualize the performance of the 23 strategies that are left out of the estimation in panel B of Figure 3. We find that these 23 long-short strategies have an average IPCA alphas of about 10 basis points, relative to an average raw return of 60 basis points.

Thus, comparing the reduction in alpha relative to raw return across panels A and B of Figure 3, we find that importance of the characteristics is better gauged by statistical significance than by the magnitude of the statistics reported in Table 5.

5.2.3 Removing ex post characteristics. Since we use characteristics that other papers have advanced, one concern is that the success of the IPCA model is because we already know that those characteristics work. The analysis in the previous section aids partly at addressing this issue. A related concern is that McLean and Pontiff (2016) show deterioration in strategy performance after publication. Thus, one might wonder whether the IPCA model is successful at explaining option returns simply because option return predictability has become weaker.

We address these issues by focusing on the 10 stock characteristics in Zhan et al. (2022) (numbered A5.9 to A5.18 in Appendix A). We focus on these characteristics because (a) they are all grouped in one paper, (b) the paper was the last one published in a top finance journal, and (c) the characteristics contain stock information that, at first glance, may not be related to the main drivers of delta-hedged option returns (all of them do generate significant long-short returns as detailed in Table 1). We remove these 10 characteristics from the estimation exercise and reestimate a restricted IPCA model with the remaining 36 characteristics. We visualize the results in panel C of Figure 3. Only one long-short portfolio has a significant alpha (after controlling for MHT). Thus, similar to the results in previous two subsections, the IPCA model is capable of explaining the returns of trading strategies that are based on characteristics that are not included in the estimation.

This particular setup also allows us to also partially address the impact of publication bias. The sample period in Zhan et al. (2022) is January 1996 to April 2016. We refer to this period as the in-sample period and the period between May 2016 and December 2012 as the out-of-sample period. We report returns and restricted IPCA alphas in Table 6 for the full sample and each of the two subsamples.

The top panel of Table 6 shows raw returns across the three sample periods. We see that the returns in the out-of-sample period (average 1.43% across 10 characteristics) are larger than those in the in-sample period (average 0.61%

Table 6
Pricing the 10 portfolios in Zhan et al. (2022)

	Full sample 1996-2022		In-sample 1996-2016/5		Out-of-sample 2016/6-2022	
	Raw returns					
Stock price	1.73	(16.06)	1.55	(14.14)	2.27	(8.30)
CashFlowVar	0.72	(9.10)	0.50	(6.19)	1.41	(7.42)
Cash to asset	0.87	(9.41)	0.60	(6.13)	1.69	(8.47)
AnalystDisp	0.52	(8.17)	0.50	(6.68)	0.59	(4.80)
1yr NewIss	0.43	(5.57)	0.17	(2.14)	1.25	(7.10)
5yr NewIss	0.65	(8.31)	0.36	(5.10)	1.55	(7.54)
Profit margin	1.15	(13.35)	0.95	(11.17)	1.77	(7.94)
ROE	0.82	(11.07)	0.69	(9.04)	1.20	(6.58)
ExternalFin	0.52	(6.09)	0.26	(2.78)	1.31	(8.07)
Z-score	0.68	(7.98)	0.49	(5.72)	1.24	(5.86)
	Restricted IPCA alphas					
Stock price	0.15	(1.11)	0.05	(0.39)	0.43	(1.33)
CashFlowVar	-0.12	(-1.40)	-0.26	(-3.23)	0.31	(1.32)
Cash to asset	0.05	(0.68)	-0.03	(-0.33)	0.31	(1.80)
AnalystDisp	0.15	(2.61)	0.16	(2.36)	0.13	(1.12)
1yr NewIss	-0.13	(-1.97)	-0.22	(-3.43)	0.17	(1.06)
5yr NewIss	-0.04	(-0.69)	-0.16	(-2.53)	0.32	(1.95)
Profit margin	0.02	(0.30)	-0.04	(-0.46)	0.21	(1.26)
ROE	-0.01	(-0.18)	-0.08	(-1.08)	0.20	(1.27)
ExternalFin	-0.22	(-2.97)	-0.30	(-3.75)	0.05	(0.35)
Z-score	0.30	(3.13)	0.18	(1.77)	0.69	(2.96)

We exclude the 10 characteristics in Zhan et al. (2022) from the set of covariates and reestimate the IPCA model on the remaining 36 (plus a constant). We refer to this estimated model as the “restricted IPCA.” We then compute average raw returns and alphas (both in percent per month) in three sample periods: the full sample (January 1996 to December 2022), in-sample (January 1996 to April 2016) corresponding to the sample studied in Zhan et al. (2022), and out-of-sample (from May 2016 to December 2020). *t*-statistics larger than the Benjamini and Hochberg (1995) 5% FDR adjusted threshold are reported in bold. MHT thresholds for raw returns are 2.22, 2.06, and 3.11, for each of the three periods, respectively. MHT thresholds for restricted IPCA alphas are 3.12, 3.21, and 3.29, for each of the three periods, respectively.

across 10 characteristics). In fact, for each of the 10 strategies, we find that the returns are higher out-of-sample than those in-sample.

The bottom panel of Table 6 shows that the pattern carries over to alphas: alphas in the out-of-sample period are higher than those in the in-sample period. However, none of the alphas is significant at MHT levels in either of the two subsamples (the strategy based on Z-score has a significant alpha at conventional levels in the out-of-sample period). From an economic point of view, the portion of return left unexplained by the IPCA alpha in the out-of-sample period is quite small at 20% (0.28/1.43).

Note, that our test is quite stringent. The restricted IPCA model does not use the 10 characteristics at all (neither in the publication in-sample period nor in the out-of-sample period). Therefore, the performance of the IPCA model in explaining the average realized returns, at least for the 10 characteristics of Zhan et al. (2022), cannot be attributed to post-publication bias.

We acknowledge that our approach does not completely rule out concerns that published papers are data mined. However, the robustness of results of the restricted IPCA model does indicate that the mechanical advantage of

IPCA in explaining a characteristic-based trading strategy by using the same characteristic as a conditioning variable is not that big.

5.3 IPCA factors

Next, we turn our attention to IPCA factors. Although these factors are latent, we attempt to characterize their properties. All factors (labeled F1, F2, and F3) have positive average return by construction. The average return of F1 is 2.06% per month, while the remaining two factors have average returns of 0.26% and 0.11%, respectively. The volatility of the factors goes from 1.38% for F1 to 0.87% for F3. The annualized Sharpe ratio of the first factor is the highest at 5.1. The pairwise correlations between factors are all negative, with the correlation between F1 and F2 being the most negative at -27% .

Factors that can price all assets with zero alpha are also the factors that can be combined to obtain the maximum Sharpe ratio (Gibbons, Ross, and Shanken 1989). We calculate Sharpe ratios of the tangency portfolio obtained by combining the three IPCA factors both in-sample and out-of-sample. The in-sample tangency portfolio calculation uses the sample average and the sample covariance matrix for portfolio weights. The out-of-sample calculation follows the same procedure as described in Section 3.2. In every month t during the out-of-sample period, we calculate the portfolio weights, $\hat{\theta}_t$, using sample moments calculated with data through t . The tangency portfolio return at time $t+1$ is then given by weights $\hat{\theta}_t$ and out-of-sample IPCA factor return, $\hat{F}_{t+1,t}$. We find that the IPCA tangency portfolio Sharpe ratio is 5.8 in-sample and only slightly lower at 4.4 out-of-sample. For comparison, the annualized Sharpe ratio of the equal-weighted delta-hedged option market portfolio is 1.5 in our sample period and about 0.8 when positions are weighted by dollar open interest. Recall that our estimation method uses gross returns to estimate IPCA factors. Armed with the portfolio weights in managed portfolios, we also calculate a tangency portfolio adjusting returns for transaction costs. We find that the Sharpe ratio for the tangency portfolio using net returns is 4.1 in-sample and 2.7 out-of-sample, and thus still economically large. Thus, the IPCA factors span a wide region of the option-return space. At the same time, we remind readers that IPCA factors are designed to capture systematic variation in cross-section of delta-hedged option returns and, thus, are not necessarily related to factors in economic models.

5.3.1 Relation to managed portfolios. IPCA factors are a linear combination of managed portfolio returns. In each month, the factor realizations correspond to $\hat{F}_{t+1} = \left(\hat{\Gamma}'_{\beta} W_t \hat{\Gamma}_{\beta} \right)^{-1} \hat{\Gamma}'_{\beta} X_{t+1} = \hat{\theta}'_t X_{t+1}$ (see Equation (4)). We plot the average contribution (through time) of each managed portfolio to the overall factor return in Figure 4.¹² We find that factor F1 is closely related to RV–IV, factor

¹² We plot the average of time-varying weights on different characteristics for each factor in Internet Appendix Figure IA4.

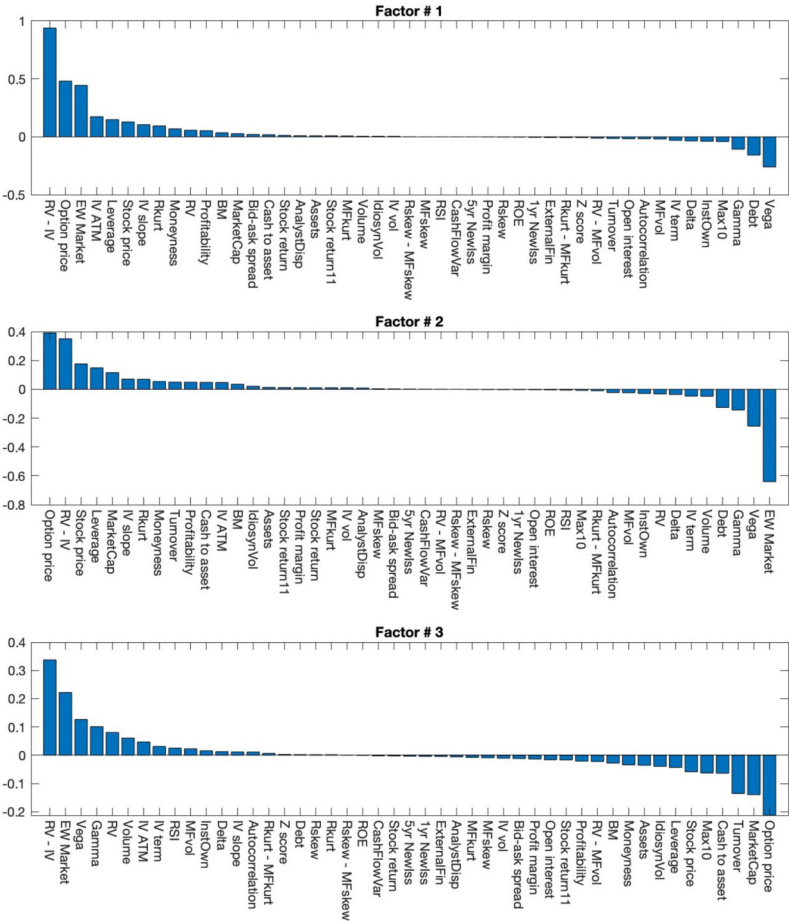


Figure 4
Managed portfolios contribution to IPCA factors

The figure shows average contribution by each managed portfolio to each of the three IPCA factors' means. The sample period is 1996 to 2022.

ALT TEXT: average contribution by each managed portfolio to each of the three IPCA factors' means shows that factor F1 is closely related to RV–IV, factor F2 is inversely linked to the equal weighted market portfolio (EW market), and factor F3 is close to a long-short combination of RV–IV and option price.

F2 is inversely linked to the equal weighted market portfolio (EW market), and factor F3 is close to a long-short combination of RV–IV and option price. These results further underscore the importance of RV–IV in explaining option returns.

5.3.2 Relation to economic conditions. Next, we provide a statistical interpretation of the factors. Specifically, we link the realization of the IPCA factors to a few proxies for adverse conditions. In panel A of Table 7, we

Table 7
IPCA factors and economic conditions

	A. Difference of means			B. Linear regression slope		
	F1	F2	F3	F1	F2	F3
Market return	−0.34 (−2.10)	0.35 (2.16)	0.35 (4.29)	−0.45 (−2.96)	0.50 (3.18)	0.22 (4.30)
Δ VIX	−0.59 (−3.95)	0.38 (2.32)	0.40 (4.91)	−0.55 (−5.28)	0.55 (3.64)	0.20 (4.43)
Tail risk	−0.46 (−2.80)	0.26 (1.60)	−0.34 (−2.90)	−0.42 (−4.98)	0.15 (1.61)	−0.25 (−3.88)
Vol risk	1.32 (8.68)	−0.72 (−4.39)	−0.27 (−2.60)	0.65 (5.85)	−0.82 (−11.11)	−0.21 (−2.05)
Skew risk	−0.81 (−4.92)	−0.03 (−0.20)	−0.23 (−2.11)	−0.36 (−4.75)	−0.02 (−0.37)	−0.15 (−2.85)
Kurt risk	−0.81 (−4.79)	−0.09 (−0.54)	−0.27 (−2.27)	−0.36 (−5.00)	−0.04 (−0.66)	−0.16 (−3.07)
ICRFac	−0.05 (−0.38)	0.37 (2.65)	0.01 (0.10)	−0.03 (−0.41)	0.28 (3.78)	−0.06 (−0.95)
Average gamma	−0.96 (−6.36)	−0.40 (−2.25)	−0.01 (−0.09)	−0.60 (−7.21)	−0.22 (−2.14)	−0.10 (−1.17)
Average vega	−0.49 (−2.76)	0.23 (1.27)	−0.27 (−2.44)	−0.10 (−1.05)	0.12 (1.13)	−0.20 (−3.28)

The table presents statistics that describe how the 3 IPCA factors are related to economic conditions. In panel A we report average differences in the returns of the factors for months during which economic indicators are higher than the median relative to months when they are below the median. In panel B we report the slope coefficient of univariate regressions of the factor returns on the realization of the same set of indicators. All independent variables are standardized to ensure comparability across variables. The set of economic indicators includes the return of the value-weighted stock market portfolio (Market Return); changes in the VIX index (Δ VIX); level of the tail risk measure of Kelly and Jiang (2014) (Tail risk); the difference between S&P 500 realized volatility, skewness, and kurtosis, and model-free implied volatility, skewness, and kurtosis (Vol risk, Skew risk, and Kurt risk, respectively); the intermediary capital risk factor of He, Kelly, and Manela (2017) (ICRFac); and the cross-sectional equal-weighted average of options gamma and vega measured at the beginning of the option initiation (Average gamma and Average vega). The model-free moments are computed at the beginning of the option initiation while the realized moments are computed over the remaining life of the option. We take negatives of Market return, Vol risk, Kurt risk, and ICRFac to ensure that high values correspond to worse economic conditions. The sample period is 1996 to 2022.

report average differences in the returns of the factors for months during which economic indicators are higher than the median relative to months when they are below the median.

We first consider how the IPCA factors relate to the negative aggregate shocks proposed by Karakaya (2013) and Büchner and Kelly (2022). We consider months in which the market portfolio return is lower than the median of the sample distribution (Market Return), or the change in the VIX index from the previous month is larger than the median (Δ VIX). Table 7 shows that F1 has lower realizations in states where the market return is negative or VIX increases; the average factor return in those states is lower than the other part of sample. F2 and F3 behave as a hedge against these negative shocks: The average return in these two negative states is approximately 35 bps higher than that in the rest of the sample, for F2 and F3, with the differences being highly statistically significant.

Second, because delta-hedged options have payoffs related to large movements in the underlying, we relate the factor returns to “extreme event risk,” as in Kelly and Jiang (2014). We study how our IPCA factors are linked

to the tail risk measure, and to differences between risk-neutral (model-free) variance, skewness, and kurtosis extracted from S&P 500 options with the same maturity of options in the delta-hedged option returns and their corresponding realized future counterparts. The model-free moments are computed on the strategy initiation date and the realized moments are computed over the period between the date on which option strategies are initiated and the expiration date of the same option contracts. The differences between realized and implied moments, thus, capture the return of volatility, skewness, and kurtosis swaps, respectively, à la Carr and Wu (2009) and Kozhan, Neuberger, and Schneider (2013).

As before we consider median breakpoints: tail risk measure higher than the median (Tail Risk), the difference of realized and implied volatility is lower than the median (Vol Risk), the difference of realized and implied skewness is higher than the median (Skew Risk), and the difference of realized and implied kurtosis is lower than the median (Kurt Risk). We find that F1 and F3 have lower realizations during periods when the tail risk measure is largest. F1 is also negatively related to high volatility, skewness, and kurtosis risk. F2 is negatively related to high volatility risk. F3 is instead negatively related to all three dimensions of higher moments risk.

Another possible source of variation in factor returns is the availability of trading counterparts. We examine states of the world when the intermediary capital ratio factor of He, Kelly, and Manela (2017), ICRFac, is lower than the median. With the exception of F2 which is positively related to ICRFac, we do not find much variation of the factor returns related to intermediary capital constraints. Related, a newer strand of literature discusses the temporary price impact of intermediaries' trading due to their hedging demand. For example, Baltussen et al. (2021) and Barbon and Buraschi (2022) document how negative gamma exposure creates intra-day momentum. Inspired by these papers, we consider instances of high Gamma or Vega, when the proper execution of delta-hedging might be challenging. We construct aggregate measures of Gamma and Vega by averaging option Greeks across all options in our sample. Note that since Gamma is inversely related to volatility, High Average Gamma spans different states of nature than High Δ VIX. We find that F1 and F2 are negatively related to instances when Average Gamma is high, while F1 and F3 are negatively related to instances when Average Vega is high.

The difference of means can be viewed as a nonparametric analysis of the relation between factor returns and different states of the world. As an alternative parametric analysis, we also consider linear relation between factor returns and the economic variables. In particular, we run the following regression:

$$F_t = a + b(X_t - \bar{X})/\sigma_X + e_t, \quad (10)$$

where we standardize X on the right-hand-side to ensure comparability of slope coefficients across different independent variables. We take negatives of

Market Return, Vol Risk, Kurt Risk, and ICRFac to ensure that high values correspond to worse economic conditions.

Panel B of Table 7 reports the results. Qualitatively, the results in Panel B are consistent with those in panel A: the slope coefficients have the same sign as the differences in means. From a quantitative standpoint, linearizing the relationship decreases the economic magnitude in some cases. For example, one standard deviation increase in Vol Risk is associated with a higher realization for the first factor of 65 basis points. However, the factor return is 1.32% higher when Vol Risk is above the median. Assuming that the distribution of X is normal, the truncated center of the distribution to the left and the right of the median corresponds to -0.7 and 0.7 standard deviations, respectively. Given that $0.65 \times 1.4 < 1.32$, this suggests a hockey-stick relationship between F1 and Vol Risk.¹³ A similar case arises for Skew Risk for F1 and for Average Vega for F1. Note that the slope of Average Vega for F1 is not even statistically significant, while the difference in means is. For the other variables, the magnitude of the realization of the factors that we obtain from the difference in means is relatively close to that implied by one standard deviation changes in the economic indicators.

Overall, we find that the three factors span different states of nature characterized by different dimensions of risk. As mentioned earlier, we urge caution in overinterpreting these results as indicating that factors returns represent risk premiums related to adverse conditions.

6. Conclusion

Explaining the returns from option positions is a daunting task. Delta-hedged option returns carry sizable liquidity premiums. We unify the knowledge accumulated in the extant literature about cross-sectional characteristics that predicts option returns and recent advances in asset pricing methods, and show how much of the profitability of delta-hedged option returns can be explained away by an IPCA factor model with conditional betas.

While both the latent factors and dynamic betas are necessary to explain option returns, one theme that emerges from our analysis is the importance of the characteristic RV–IV. We find not only that a strategy based on RV–IV produces high returns of 2.87% per month but also that the characteristic RV–IV is related to covariances with respect to risk factors. While the importance of a characteristic related to volatility, such as RV–IV, is not entirely surprising in explaining returns to volatility-sensitive delta-hedged call positions, what our analysis highlights is that RV–IV captures commonality in many other predictors of option returns. Setting the covariance coefficients, Γ_β , related to RV–IV to zero greatly decreases the ability of the IPCA model to

¹³ In unreported results, we find that a regression with two different slopes (to the right and to the left of the median) confirms this interpretation.

price any portfolio that is not related to RV—IV: for example, 32 strategies see an increase in alpha and that increase is 1% on average.

Our results are important because they reaffirm the idea that market efficiency is still a valid framework to think about how prices form in markets. The fact that this is true even in a market that is highly segmented and riddled with informational advantages, as is the stock option market, is quite remarkable.

Appendix A. Variable Construction

A1. Measures Related to Contract Characteristics

1. Moneyness: Moneyness defined as the ratio of strike to underlying price.
2. Bid-ask spread: Bid-ask spread of the contract on the initiation day of the strategy.
3. Open interest: Dollar open interest of the contract (i.e., open interest defined as number of contracts times the most recent option price).
4. Delta: Option delta.
5. Vega: Option vega.
6. Gamma: Option gamma.
7. Volume: Dollar option volume on the initiation day of the strategy (i.e., volume defined as number of contracts times the most recent option price).
8. Option price: Option price calculated as the midpoint of the bid and ask quotes.

A2. Measures Related to the Risk-Neutral Distribution of Returns

1. IV ATM: The ATM implied volatility extracted from the 30 days volatility surface.
2. IV slope: The difference between the OTM implied volatility (i.e., ratio of strike to underlying equal to 0.8) extracted from the 30 days volatility surface and the corresponding ATM implied volatility.
3. IV term: The difference between the ATM implied volatility extracted from the 360 days volatility surface and the corresponding implied volatility extracted from the 30 days surface as in [Vasquez \(2017\)](#).
4. IV vol: Volatility of implied volatility from the volatility surface of Option-Metrics. We use implied volatility of calls with 30 days to maturity with a delta of 0.5 and calculate the standard deviation using daily data over the last month with a minimum of 15 days.
5. MFvol: model-free implied volatility is constructed from 30 days OTM call and OTM put option prices as in [Bakshi, Kapadia, and Madan \(2003\)](#). In particular we follow the procedure described in [Hansis, Schlag, and Vilkov \(2010\)](#), whose code is available on Grigory Vilkov's web page.
6. MFskew: model-free implied skewness is constructed from 30 days OTM call and OTM put option prices as in [Bakshi, Kapadia, and Madan \(2003\)](#). In particular we follow the procedure described in [Hansis, Schlag, and Vilkov \(2010\)](#), whose code is available on Grigory Vilkov's page.
7. MFkurt: model-free implied kurtosis is constructed from 30 days OTM call and OTM put option prices as in [Bakshi, Kapadia, and Madan \(2003\)](#). In particular we follow the procedure described in [Hansis, Schlag, and Vilkov \(2010\)](#), whose code is available on Grigory Vilkov's page.

A3. Measures Related to the Physical Distribution of Returns

1. Stock return: Monthly stock return.
2. Stock return11: Stock return over the last 11 months skipping the most recent month.
3. RV: Volatility of log returns calculated using daily data over the last 12 months with a minimum of 150 observations.
4. Rskew: Skewness of log returns calculated using daily data over the last 12 months with a minimum of 150 observations.
5. Rkurt: Kurtosis of log returns calculated using daily data over the last 12 months with a minimum of 150 observations.
6. Turnover: Ratio of number of shares traded over the last month to total shares outstanding.
7. IdiosynVol: Standard deviation of residuals from [Fama and French \(1993\)](#) three-factor model. We use daily data over the last month with a minimum of 10 days.
8. Max10: Average of the 10 highest daily returns over the last 3 months, following [Bali, Cakici, and Whitelaw \(2011\)](#).
9. Autocorrelation: Autocorrelation of returns calculated using daily data over the last 6 months with a minimum of 100 observations, following [Jeon, Kan, and Li \(2024\)](#).

A4. Measures Related to Differences between the Physical and Risk-Neutral Distribution of Returns

1. RV–IV: difference between realized and option implied volatility.
2. RV–MFvol: difference between realized and model-free volatility.
3. Rskew–MFskew: difference between realized and model-free skewness.
4. Rkurt–MFkurt: difference between realized and model-free kurtosis.

A5. Stock-Level Variables

1. BM: Book-to-market ratio is calculated as the ratio of book value of equity, calculated as in [Fama and French \(1992\)](#) to the current market value of equity.
2. Profitability: Profitability, as in [Novy-Marx \(2013\)](#), is calculated as the ratio of gross profits (GP) to total assets (AT).
3. InstOwn: Institutional ownership percentage (`instown_perc`) from Thomson Reuters 13f holdings.
4. MarketCap: Market capitalization.
5. RSI: The ratio of shares that are sold short (`shortintadj` in Compustat short interest file) to the total shares outstanding, as in [Ramachandran and Tayal \(2021\)](#).
6. Assets: Total assets (AT).
7. Debt: Total debt defined as the sum of long-term debt (`DLTT`) and debt in current liabilities (`DLC`).
8. Leverage: Ratio of financial debt to assets.
9. CashFlowVar: Cash flow variance, as in [Haugen and Baker \(1996\)](#), is computed as the variance of the monthly ratio of cash flow to market value of equity over the last 60 months. Cash flow is net income (`IB`) plus depreciation and amortization (`DP`).

10. Cash to asset: The cash-to-assets ratio, as in [Palazzo \(2012\)](#), is the value of corporate cash holdings (CHE) over the value of the firm's total assets (AT).
11. AnalystDisp: Analyst earnings forecast dispersion, as in [Diether, Malloy, and Scherbina \(2002\)](#), computed as the standard deviation (`stddev`) of annual earnings-per-share forecasts (EPS1) scaled by the absolute value of the average outstanding forecasts (`meanest`).
12. 1yr NewIss: 1-year new issues, as in [Pontiff and Woodgate \(2008\)](#), measured as the change in log of shares outstanding (`shrout`) from 11 months ago. Shares outstanding are adjusted for splits using the cumulative adjustment factor to adjust shares (`cfacshr`).
13. 5yr NewIss: 5-year new issues, as in [Daniel and Titman \(2006\)](#), measured as 5-year real change in log of number of shares outstanding.
14. Profit margin: Profit margin, as in [Soliman \(2008\)](#), calculated as earnings before interest and tax (OIADP) scaled by revenues (SALE).
15. Stock price: The logarithm of stock price at the end of last month, as in [Blume and Husic \(1972\)](#).
16. ROE: Profitability, as in [Fama and French \(2006\)](#), calculated as earnings divided by book equity, in which earnings is defined as income before extraordinary items (IB), and book equity is calculated as in [Fama and French \(1992\)](#).
17. ExternalFin: Total external financing, as in [Bradshaw, Richardson, and Sloan \(2006\)](#), calculated as net share issuance plus net debt issuance minus cash dividends, scaled by total assets (AT). Net share issuance is computed as net cash received from the sale (and/or purchase) of common and preferred stock (SSTK less PRSTKC) less cash dividends paid (DV). Net debt issuance represents net cash received from the issuance (and/or reduction) of debt (DLTIS less DLTR plus DLCCH). DLCCH is set to zero if missing.
18. Z-score: Z-Score, as in [Dichev \(1998\)](#) constructed as $[1.2 \times (\text{Working Capital}/\text{Assets}) + 1.4 \times (\text{Retained Earnings}/\text{Assets}) + 3.3 \times (\text{EBIT}/\text{Assets}) + 0.6 \times (\text{Market Value of Equity}/\text{Book Value of Total Liabilities}) + (\text{Revenues}/\text{Assets})]$. Working capital is the difference between ACT and LCT, Retained earnings are RE, EBIT is OIADP, Revenues are SALE, Total Liabilities are LT, Assets are AT, and the market value of equity is the market capitalization at fiscal year-end.

Appendix B. IPCA Alpha Standard Error Calculation

Consider a portfolio with returns R_{t+1} and characteristics Z_t . The alpha of this portfolio is defined in Equation (9) as $\hat{\alpha} = \sum_t (R_{t+1} - Z_t' \hat{\Gamma}_\beta \hat{F}_{t+1}) / T$. Define $\hat{\alpha}_{t+1} = R_{t+1} - Z_t' \hat{\Gamma}_\beta \hat{F}_{t+1}$. Then we have:

$$\begin{aligned} \hat{\alpha}_{t+1} &= R_{t+1} - Z_t' \hat{\Gamma}_\beta \hat{F}_{t+1} \\ &= \left(R_{t+1} - Z_t' \bar{\Gamma}_\beta \hat{F}_{t+1} \right) - Z_t' E_\beta \hat{F}_{t+1}, \end{aligned} \quad (\text{B1})$$

where we have accounted for the fact that the $L \times K$ matrix $\hat{\Gamma}_\beta$ has estimation error specified by E_β . The standard error of $\hat{\alpha}$ needs to correct for the estimation error in $\hat{\Gamma}_\beta$ and $\hat{F}$. Obtaining an analytical expression for the variance of $\hat{\alpha}$ is a difficult, if not impossible, task. We make some simplifying conservative assumptions below to make some headway in this issue.

First, ignoring the estimation error in the latent factor estimates, we have:

$$\begin{aligned}\text{var}(\hat{\alpha}_{t+1}) &= \text{var}(R_{t+1} - Z_t' \bar{\Gamma}_\beta F_{t+1}) + \text{var}(Z_t' E_\beta F_{t+1}) \\ &\quad - 2\text{cov}(R_{t+1} - Z_t' \bar{\Gamma}_\beta F_{t+1}, Z_t' E_\beta F_{t+1}) \\ &= \text{var}(R_{t+1} - Z_t' \bar{\Gamma}_\beta F_{t+1}) + \text{var}(Z_t' E_\beta F_{t+1}).\end{aligned}\quad (\text{B2})$$

where, in the second line, we have made use of the second simplifying assumption that E_β is uncorrelated with R , Z , and F .

Note that $Z_t' E_\beta F_{t+1} = \sum_{l=1}^L Z_{t,l} E'_{\beta,l} F_{t+1}$, where $Z_{t,l}$ is the l th characteristic at time t and $E_{\beta,l}$ is $K \times 1$ vector of estimation errors in Γ_β estimates for the l th characteristics. We now make the third simplifying assumption that the estimation error in betas across different characteristics are uncorrelated. We then have:

$$\text{var}(Z_t' E_\beta F_{t+1}) = \sum_{l=1}^L \text{var}(Z_{t,l} E'_{\beta,l} F_{t+1}). \quad (\text{B3})$$

Since $Z_{t,l} E'_{\beta,l} F_{t+1} = \sum_{k=1}^K Z_{t,l} E_{\beta,l,k} F_{t+1,k}$, we have:

$$\begin{aligned}\text{var}(Z_{t,l} E_{\beta,l,k} F_{t+1,k}) &= \text{var}(Z_{t,l} F_{t+1,k}) \times [\text{var}(E_{\beta,l,k}) + \overline{Z_{t,l} F_{t+1,k}}^2] \\ \text{cov}(Z_{t,l} E_{\beta,l,k_1} F_{t+1,k_1}, Z_{t,l} E_{\beta,l,k_2} F_{t+1,k_2}) &= \text{cov}(E_{\beta,l,k_1}, E_{\beta,l,k_2}) \times \overline{Z_{t,l}^2 F_{t+1,k_1} F_{t+1,k_2}}.\end{aligned}\quad (\text{B4})$$

Combining Equations (B2) to (B4), we get:

$$\begin{aligned}\text{var}(\hat{\alpha}_{t+1}) &= \text{var}(R_{t+1} - Z_t' \bar{\Gamma}_\beta F_{t+1}) \\ &\quad + \sum_{l=1}^L \left\{ \sum_{k=1}^K \text{var}(Z_{t,l} F_{t+1,k}) \times [\text{var}(E_{\beta,l,k}) + \overline{Z_{t,l} F_{t+1,k}}^2] \right. \\ &\quad \left. + \sum_{k_1=1}^K \sum_{k_2=1, k_2 \neq k_1}^K \text{cov}(E_{\beta,l,k_1}, E_{\beta,l,k_2}) \times \overline{Z_{t,l}^2 F_{t+1,k_1} F_{t+1,k_2}} \right\}.\end{aligned}\quad (\text{B5})$$

Finally, the standard error of $\hat{\alpha}$ is given as usual by $\sqrt{\text{var}(\hat{\alpha}_{t+1})/T}$. Equation (B5) requires variances of covariances of E_β . Fortunately, these are already available via our bootstrap procedure from Section 3.1. The other quantities in Equation (B5) are replaced by their sample counterparts.

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